



FACULTY OF
APPLIED SCIENCES
SCHOOL OF ENGINEERING SCIENCE

LithoNow Maskless Lithography System Design Specifications

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To: Professor Michael Hegedus
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Dear Dr. Hegedus and Capstone evaluation team,

Please find enclosed the design specification document for LithoNow, a capstone project for a DIY lithography system that could be used in research by Prof. Michael Adachi and his Nano Device Fab Group in the School of Engineering Science.

This document outlines the design of the system's six main subsystems, prototype results, safety strategy, testing plan, and engineering standards for the capstone project. We hope to reflect the current status and lessons learned in integrating the electrical, electronic, control, mechanical and software components all together. We also outline key priorities for Capstone B, including stage motion accuracy and beam focusing. We respectfully submit this document for your review and evaluation.

Regards,

The LithoNow Team
Company #12

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Introduction

LithoNow is a benchtop maskless lithography system designed to fabricate micro-scale semiconductor patterns directly on photoresist-coated silicon wafers. The primary objective is to enable rapid prototyping of semiconductor devices without requiring photomasks, which has long been in demand functionality in the ENSC cleanroom. The machine can operate outside of a cleanroom, as it keeps wafers clean using a filtration system while still being affordable, very high in demand for small labs and companies who are unable to operate an ISO-certified cleanroom.

The system will support research conducted by Professor Michael Adachi's Nano Device Fabrication group.

The device focuses on producing the Schottky diode evaluation chip provided by the research group. The smallest feature in the design is approximately 3 μm . The system must therefore achieve feature sizes between 3 μm –5 μm during the engineering prototype stage.

The system must also operate outside of a cleanroom environment while maintaining ISO Class 5 cleanliness locally around the wafer.

The design specifications presented here describe the full system design needed to satisfy the requirements specification document. Each subsystem references the associated requirement IDs defined in the requirements specification document.

System Architecture

The LithoNow system contains the following six subsystems:

- Laser & optical system, fabrication (Vitaliy)
- Power, laser safety, electrical interlock (Armin)
- Environmental filtration enclosure (Tony)
- XYZ stage control system electronics & firmware (Eric)
- XYZ stage mechanical design (Steven)
- Software/GUI integration & laser writing algorithm (Justin)

Each subsystem implements requirements from the requirements specification.

Subsystem	Function	Requirement IDs
Laser optics	expose photoresist	G1.0, E4.1, E4.2
XYZ Stage	wafer positioning	M5.1, M5.2
Camera alignment	alignment marks	M4.6, S4.5
Air filtration	contamination control	M7.1
Software	.GDS processing	S0, S1
Safety enclosure	laser containment	G1.1, G1.2

Table 1: Subsystem summary

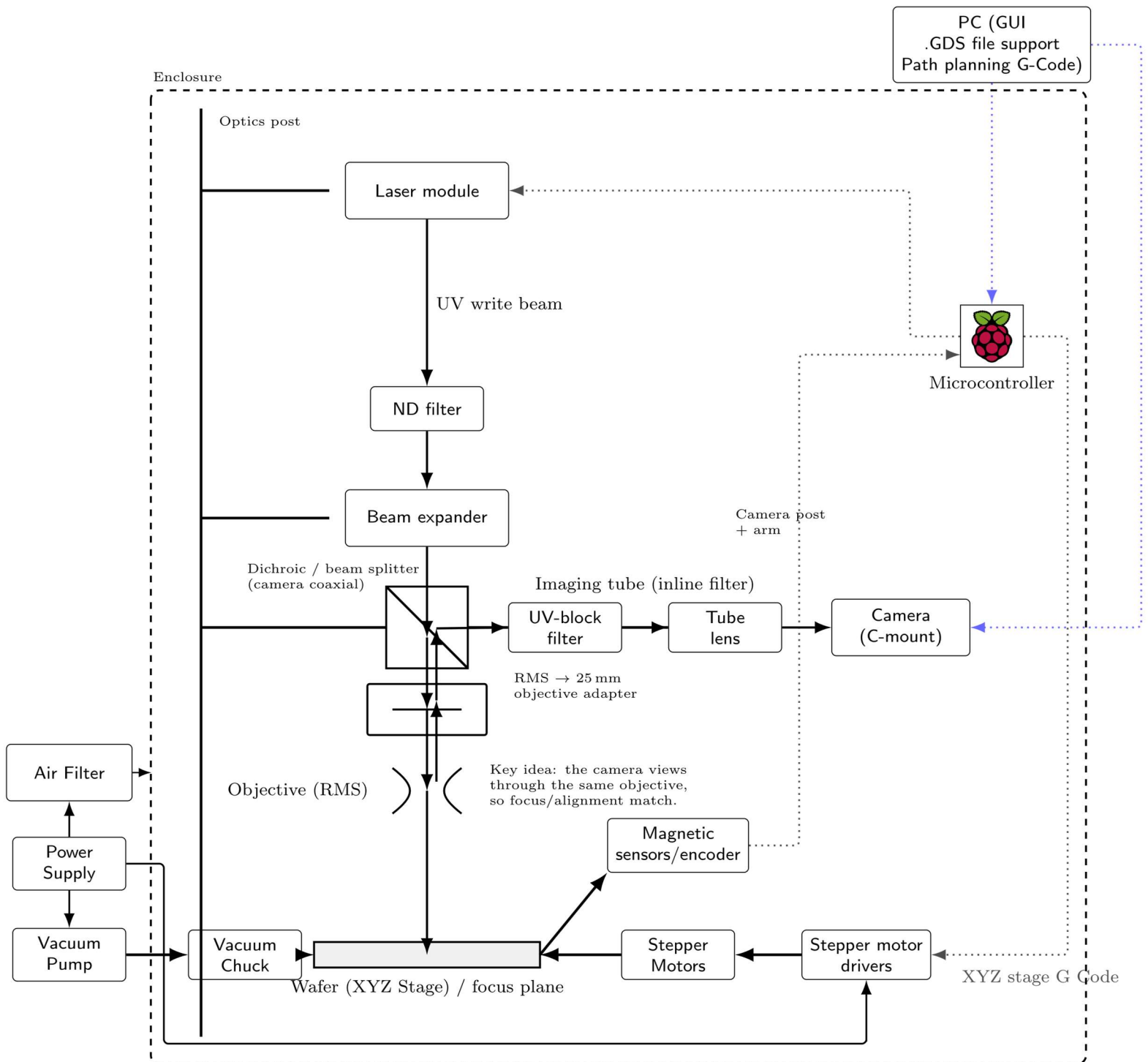


Figure 1: Block diagram overview of the entire system whose individual components will be discussed throughout this document. A full CAD rendering can be found on Appendix D.

Optical System Design

The optical system implements lithography requirements G0, E4.1, and E4.2.

Laser Selection

The system uses a Class 2 1mW 405 nm diode laser. This wavelength satisfies requirement [1.0], [G0] which specifies compatibility with S1813 photoresist (requires 400-700nm).

The laser is controlled using TTL modulation implemented by the Raspberry Pi Pico microcontroller. In practice, this is relatively simple: the laser module is powered from its DC supply pins while a Pico digital output provides on/off gating through the TTL control input, enabling software-synchronized exposure timing with stage motion.

Commercial 405 nm laser modules of this class provide dedicated TTL control connections and DC power leads, which supports straightforward microcontroller integration.

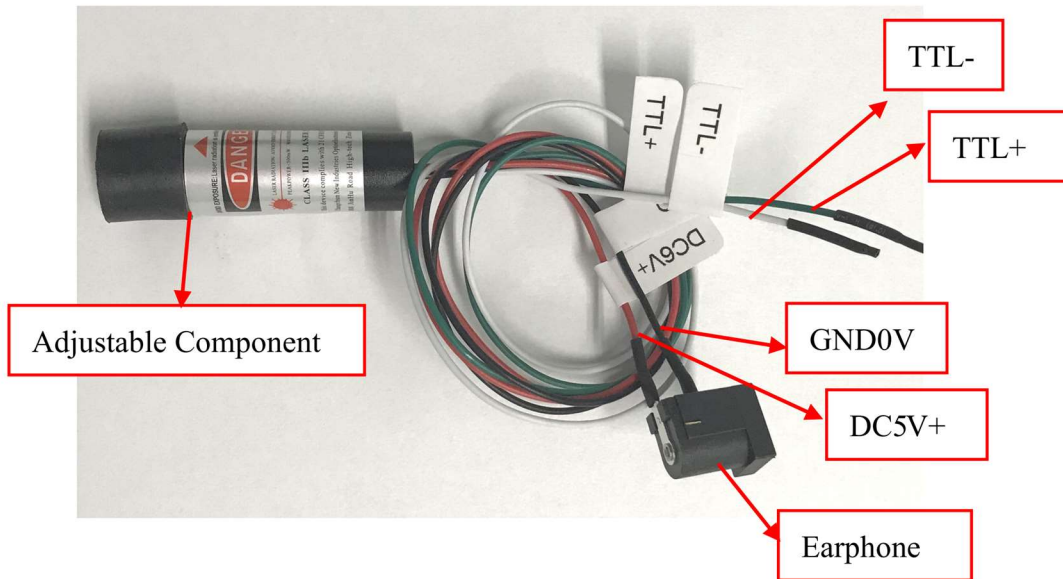


Figure 2: Annotated wiring view of the 405 nm laser module showing TTL and DC leads used for Pico-based modulation control.

This design fulfills (G0 [1.0]: S1813 exposure compatibility, E0: TTL laser control).

Diffraction Limited Spot Size

The theoretical diffraction-limited laser dot diameter is

$$d = \frac{1.22\lambda}{NA}$$

For

$$\lambda = 405 \text{ nm}$$

$$NA = 0.95$$

$$d \approx 520 \text{ nm}$$

Practical spot size if considering non-ideal conditions:

$$d_{real} \approx 0.6 - 0.8 \mu\text{m}$$

We do have access to lenses with lower NA which could produce a bigger dot or one closer to the 3µm feature size goal we need for the Schottky diode design. These laser considerations satisfy the design target for requirement E4.2 (3 µm–5 µm feature capability).

Depth of Focus

$$\text{DOF} = \frac{\lambda}{2NA^2}$$

$$\text{DOF} \approx 225 \text{ nm}$$

Because the depth of focus is small, a vacuum chuck is required to maintain wafer flatness.

This design fulfills (M4.1: wafer flatness, B M3: vacuum chuck). **Optical Layout**

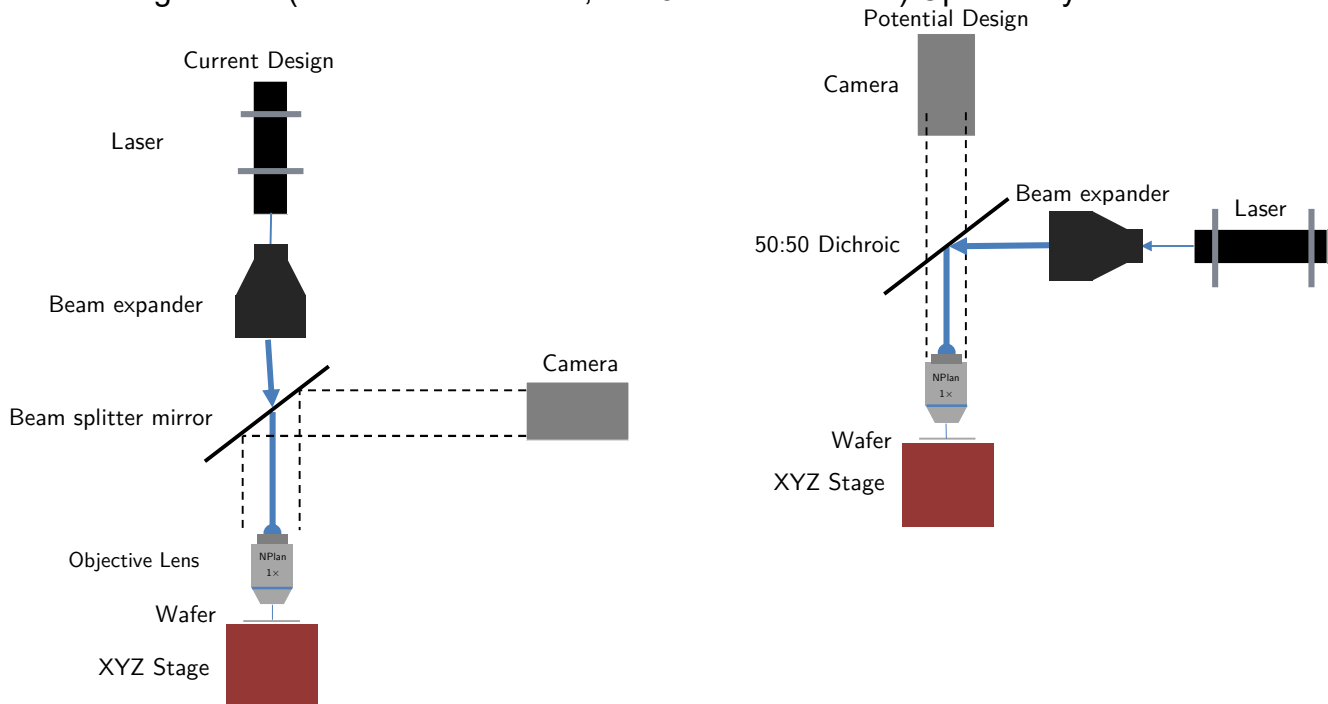


Figure 2: Left: Current optical layout design with a beam splitter configuration. Right: Backup optical layout design with a dichroic mirror configuration.

The optical system consists of:

- 405nm 1mW UV laser module
- beam expander
- beam splitter (we may switch to a dichroic mirror)
- microscope objective lens
- camera for wafer mark alignment

This optical layout fulfills (M4.6: camera sees laser path, S4.5: camera video to GUI, M4.2: magnification switching). There is often concern when using 405nm lasers in lithography being that it has a non-uniform gaussian intensity and often has an elliptical dot shape. This is due to fundamental physics of lasers of this wavelength and a higher quality 405nm module won't change it.

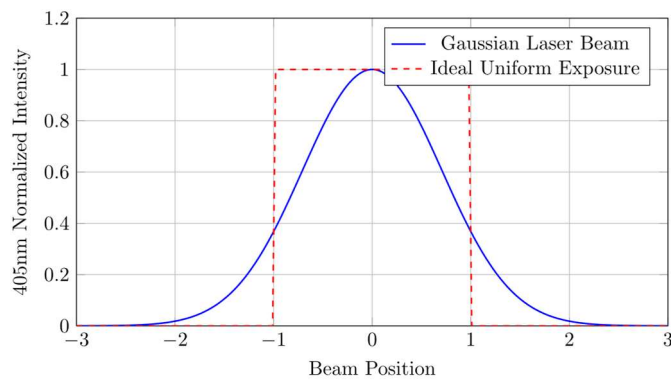


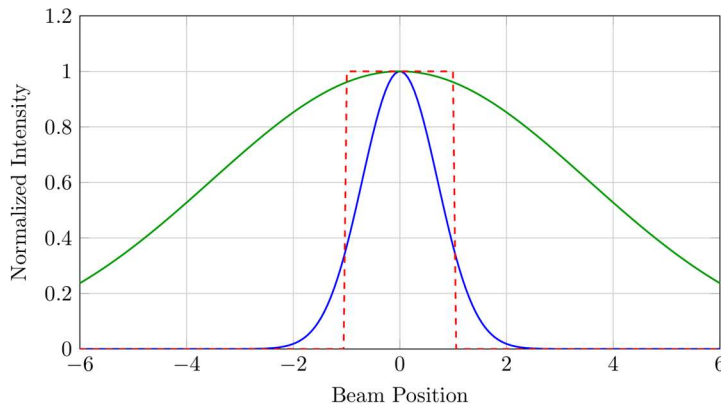
Figure 10a: 405nm laser intensity distribution



Objective lens aperture
(not fully filled)

Figure 10b: unfilled objective lens aperture

In order to combat this issue, we've added in a beam expander (rated for 5 $\times$ expansion). This turns the 3.5mm diameter laser dot into 10.5mm diameter dot. This will actually overfill the aperture of the objective lens which will force a circular dot rather than an elliptical shape. Moreover, as seen in Figure 10a, it will create a more uniform intensity on the exposed area.



— Original Gaussian Beam — 5 $\times$ Beam Expanded Gaussian - - - Ideal Uniform Exposure

Figure 11a: 405nm intensity distribution with 5 $\times$ beam expansion



Objective lens aperture
(overfilled with 5 $\times$ dot)

Figure 11b: overfilled objective lens aperture

Another possible design choice is using a beam splitter mirror versus a dichroic mirror. Although a beam splitter mirror was provided by Michael Adachi, it can result in a 50% reduction in the laser power, giving us only ~ 0.4 - 0.5 mW to work with. This can double the time needed for exposure. Or may reduce the laser power even further than we think. Another option is to use a dichroic mirror, which instead of having pass through a glass medium, instead just bounces (reflects) off a mirror. This preserves the power to at least 0.9 mW. This dichroic mirror configuration should also provide a better image for the camera, where with a beam splitter the camera suffers a 50% contrast reduction and could prove difficult to focus in on an image. This is all of course still up to experimentation.

Vacuum Chuck for Focus Stability (Optics-Critical)

The vacuum chuck is included in the laser/optics domain because its primary function is optical-focus stability at the wafer surface. With a 405 nm writing laser and high-NA focusing, the depth of focus is on the order of 200 nm (calculated above as approximately 225 nm). Even small wafer bow or local lift can move photoresist out of focus, which directly degrades feature quality.

For this reason, Vitaliy is assigned this subsystem interface: maintaining wafer flatness under vacuum is directly tied to beam focusing performance, spot control, and repeatable exposure quality.

The concept uses a hollow vacuum stage connected to a pump [Requirement: M3]. This is a Capstone B optimization and will be a part of our improved prototype. That's why in the following drawing you will see a "theta stage"; this is a Capstone B consideration (in the future when we'll need to do rotational adjustment, not just XYZ).

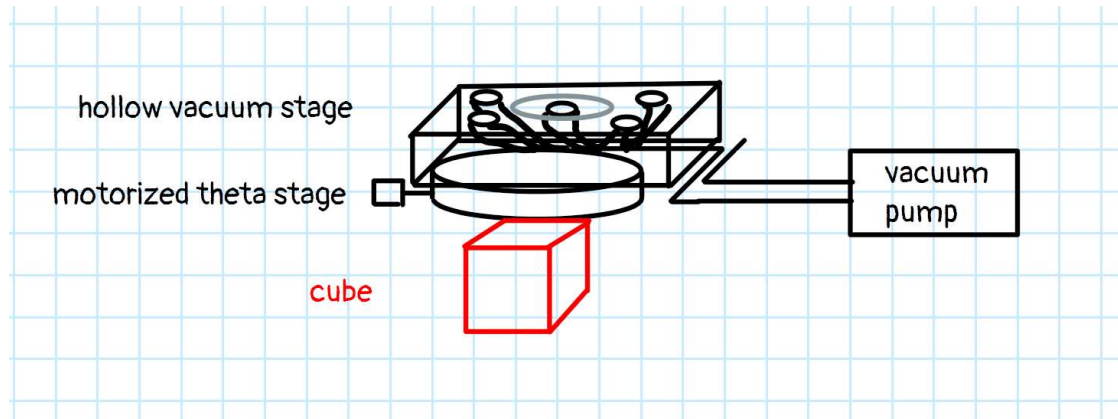


Figure 3: Concept sketch of the vacuum chuck assembly with pump connection and theta-stage integration for wafer holding and focus stability. (Responsible: Vitaliy – Laser & Optics Beam Focusing)

Vacuum sealing is achieved with an O-ring seated in a non-circular/racetrack-style groove that is typical in fab equipment requiring a vacuum seal [6]. We will use CNC with aluminum stock which we found from decommissioned fab equipment from 4D Labs.

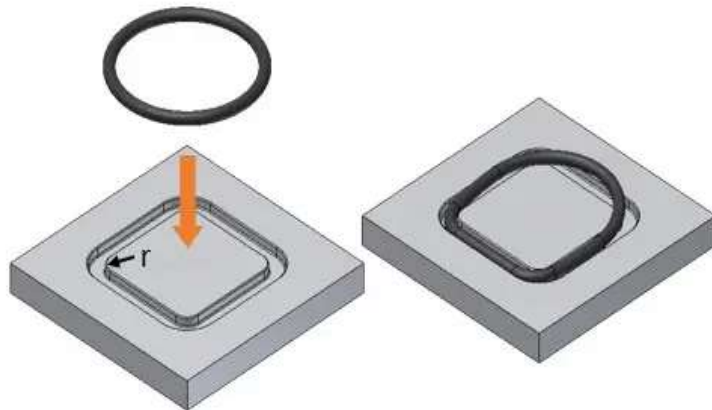


Figure 4: O-ring and groove concept for forming a stable vacuum seal in a non-circular groove pattern.

During assembly, bolt preload compresses the O-ring to create sealing pressure. Proper compression control is important to avoid under-sealing or gasket damage.

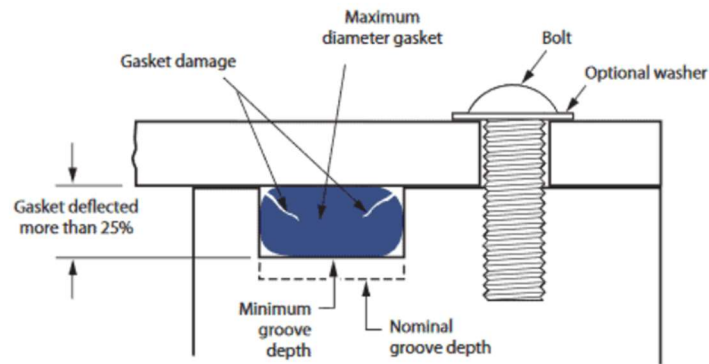
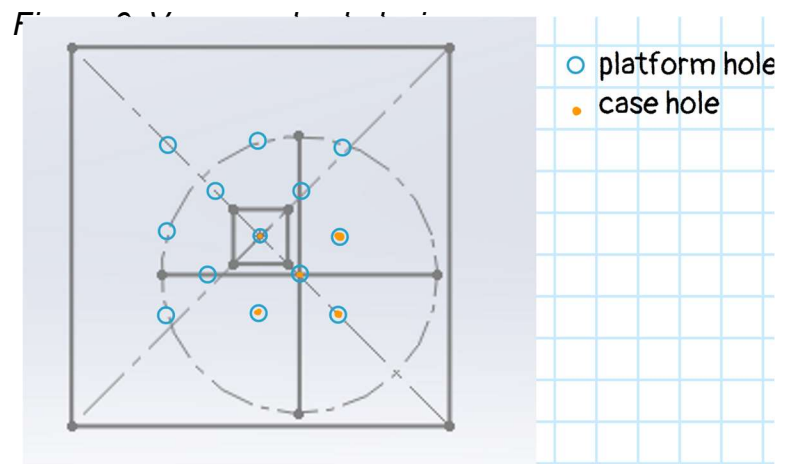
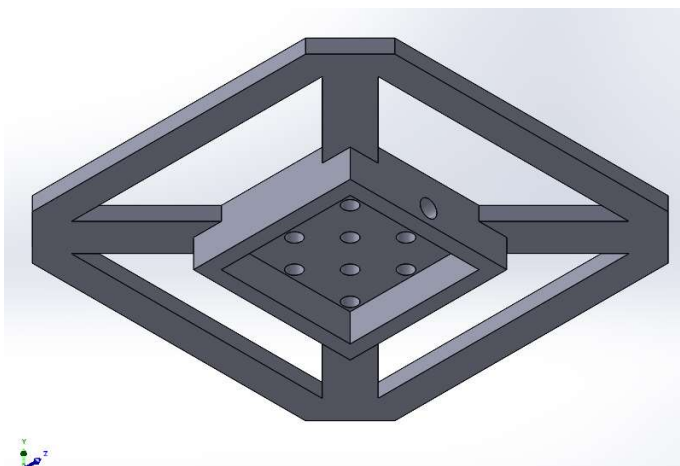


Figure 5: O-ring compression concept under bolt preload, illustrating groove-depth and gasket-deflection considerations for vacuum sealing [6].

As for the vacuum holes themselves, we considered a number of designs. Initially, we were going to make many equally sized holes. And then we found out that in order to have more uniform spreading of the pulling force and more uniform flatness, we require grooves with a small depth to them. Circular grooves in particular work best in achieving a uniform pulling force on the wafer.



There is also a problem if you have the same hole size at the edge of the vacuum chuck. The edges of wafer is very sensitive and if you have a strong pull force on the edges, it will cause

A circular diagram with three concentric circles. The outermost circle is thick black, while the two inner circles are thin gray. Six radial lines, consisting of one thick black line and five thin gray lines, divide the circles into sectors. A central cluster of points is located at the origin. Additional points are placed on the radial lines: one point on the thick black line at the intersection with the middle gray circle, and one point on each of the five thin gray lines at the intersection with the outermost thick black circle.

- Center vacuum hole
- Quadrant vacuum hole
- Edge vacuum hole
- Vacuum groove

Technical drawing of a Vacuum-Plenum assembly. The drawing includes the following views and dimensions:

- Left View:** Shows a vertical profile with a central hole of diameter $\phi 8.43$ and a depth of 13.00. The hole is labeled "1/8 NPT". The overall width is 17.50.
- Top View:** Shows a square footprint with an outer dimension of 102.00 and an inner dimension of 82.00. It features an "O-ring groove" and a chamfer of $\phi 3.40$. The distance from the center to the outer edge is 24.38.
- Front View:** Shows the side profile with a total height of 17.50. The top flange has a thickness of 14.23, and the bottom flange has a thickness of 13.00. The central opening has a width of 81.00.
- Bottom View:** Shows the underside of the plenum with a central square feature of 81.00 by 81.00. It includes chamfers of $\phi 3.40$ and a distance of 24.38 from the center to the outer edge.
- Isometric Views:** Three isometric views showing the 3D structure. Key features include:
 - O-ring Depth: 2.00
 - TRUE R1.00 (fillets at corners)
 - TRUE R0.60 (fillets on the side flange)
 - TRUE R4.74 (fillets on the side flange)
 - TRUE R1.00 (fillets at the bottom corners)

Callouts and Tolerances:

- M3x0.5 $\phi 3.40$ ∇ 21.00 (Thread specification)
- $\phi 8.43$ ∇ 13.00 (Hole specification)
- $\phi 3.40$ (Chamfer specification)

Title Block:

UNLESS OTHERWISE SPECIFIED:
 DIMENSIONS ARE IN MILLIMETERS
 SURFACE FINISH:
 TOLERANCES:
 LINEAR:
 ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

NAME: Vitaliy Vekshyev

SIGNATURE: [Signature]

DATE: 26/02/14

CHCKD:

APPVTD:

MFG:

QA:

MATERIAL:

WEIGHT:

TITLE: Vacuum-Plenum

DWG NO: A3

SCALE: 1:1

SHEET 1 OF 1

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This optics-driven chuck design fulfills (M4.1: wafer flatness, B M3: vacuum chuck, E4.1: optical spot performance stability).

Mechanical Stage Design

Motor horn and rod mechanism used to push the XYZ platform during stage actuation, this is how the movement of the wafer is achieved while the laser is in a fixed position [Requirement: M0]. The mechanism consists of NEMA 17 stepper motors, a motor horn with magnets, and rods with ball joints with the end effector being a block that acts as the platform. Electronics would just be the encoder that takes input from the row of motors on the motor horn.

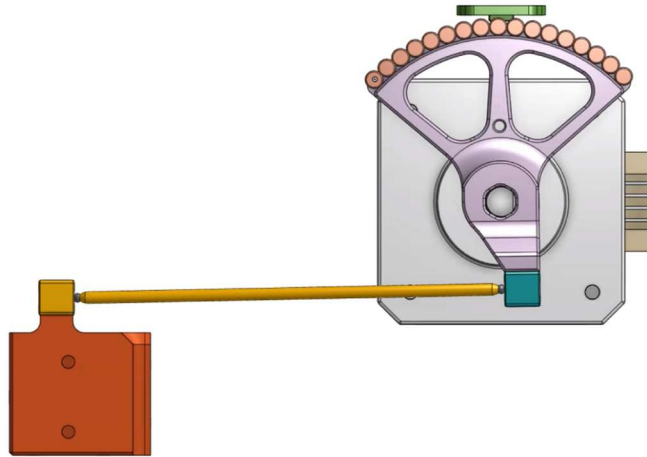


Figure 11: Mechanism of moving the end effector

The mechanical stage design is based on the OpenMicroManipulatorStepper [2] architecture. The initial design consists of movement in the X, Y, and Z axis. Most importantly the X and Y axis will be used the most to print the circuits. The Z stage has the benefit of elevating the platform to allow the camera to focus.

Subsystem	Function	Requirement IDs
Mechanical	<i>Accuracy of XYZ stage to be 7um</i>	M5.1, M5.2, M0.1

The XYZ stage design fulfills (M5.1: stage accuracy, M5.2: improved precision, M0.1: continuous operation).

Motor Mount Platform

The motor mount platform's primary function is to hold the NEMA 17 motors firmly in place with little to no play as small tolerances could cascade into large inaccuracies and decrease the lasers precision. Currently the mount material is PETG as it is low cost and prints with decent resolution. Some potential drawbacks of PETG are the thermal rating as it is rated up to 70° C before it starts to deform and bend causing potential errors or thermal drift.

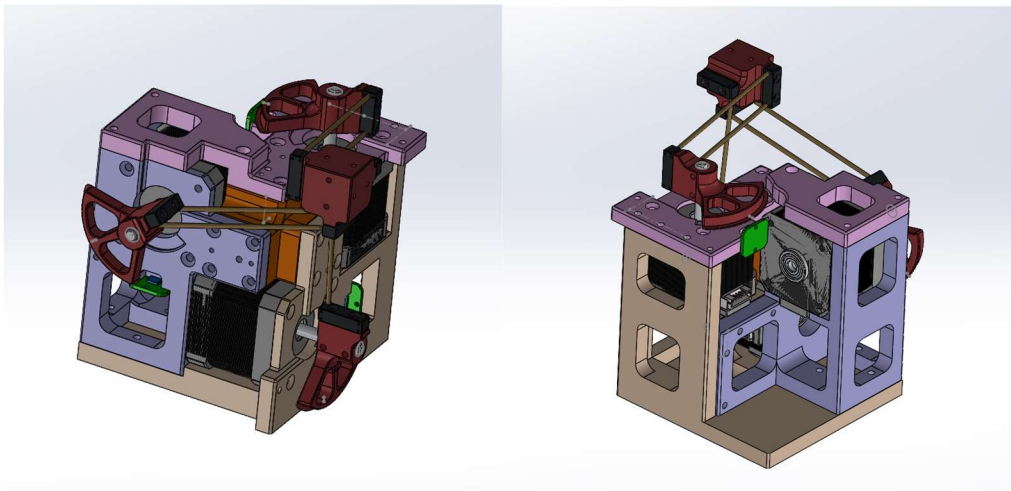


Figure 12: Motor mount platform

The mounts are modified from the opensource GitHub into a three-piece system that uses M3 screws into each other, creating an interlock that functions as a clamp to hold each of the motors into each other. With this kind of design even with some thermal deformation, the motors will still be secured in place. There is plenty of space on the floor of the design to drill holes into so that it can be screw mounted onto T-slot aluminum of the enclosure.

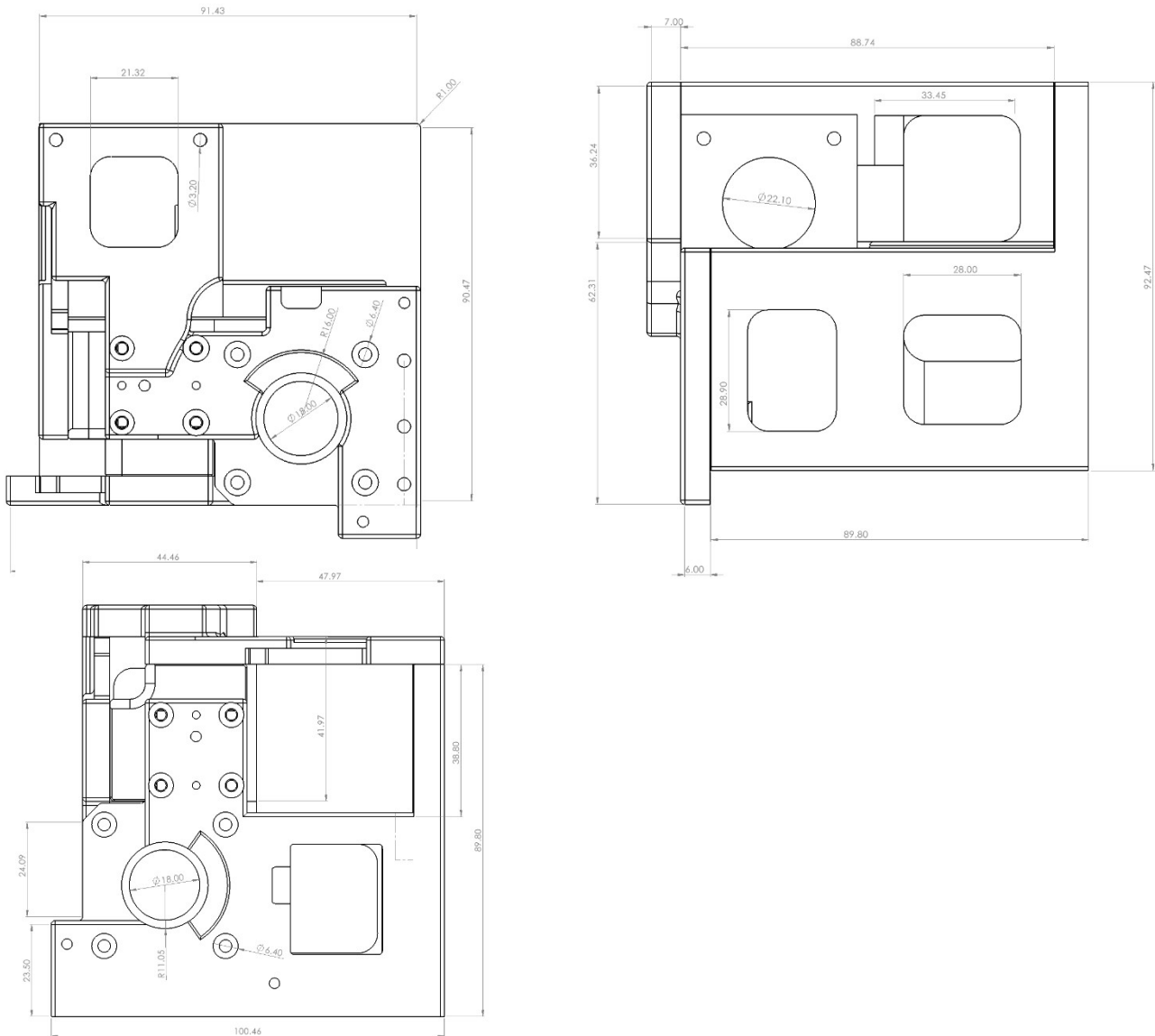


Figure 13: Motor mount sketch

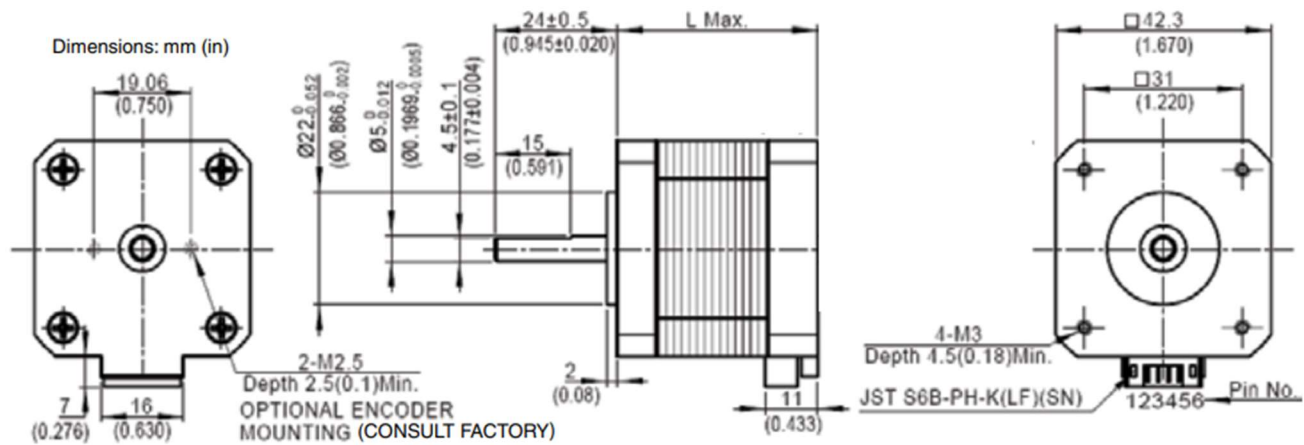


Figure 14: NEMA 17 Motor Dimensions [12]

Item	Specification
Phases	2
Steps/Revolution	200
Step Accuracy	±5%
Shaft Load	200 hours at 1000 RPM
Axial Push Load	25 N
Axial Pull Load	65 N
Radial Load	29 N
IP Rating	40
Operating Temp	-20° C to +40° C
Insulation Resistance	100 MegOhms
Rated Current	2 A
Weight	0.28 kg
Voltage	12 V

Table 2: NEMA 17 Specifications

Stage Construction

Each axis consists of:

- NEMA17 stepper motor
- magnetic encoder
- direct drive linkage
- rigid aluminum mounting

Closed-loop control corrects mechanical error.

This architecture avoids backlash associated with gear systems.

Encoder Choice Rationale and Trade-offs

The stage uses magnetic encoder feedback with stepper actuation to increase precision without relying on high-ratio mechanical gears. This reduces tolerance stacking and backlash from physical gear trains while preserving a compact architecture. Tang and Yu showed that adding an AS5600 magnetic encoder enables practical closed-loop stepper control with low average sampling error (0.3258% over 1.8° – 360°), supporting this choice for low-cost precision motion control [3]. We have considered optical encoders but could not find optical encoders that would have the same resolution as the MT6835, at least within our budget. We found the MT6835 to be especially good value for our project for being affordable as well as a gamechanger for homebrew precision motion with its 21-bit precision and latency of 10 μ s, ideal for fast and precise closed-loop responses [6].

We also use a rod-and-ball linkage held in tension by a stretchable string and a 3D-printed clip. The preload keeps mating features seated and reduces mechanical play during direction changes.

A key geometric trade-off is rod radius from the axis of rotation. If the rod is positioned closer to the center of rotation, we obtain more precise resolution and smoother movement; however, we trade off the total area that can be covered in a single lithography run. We prioritize precision for lithography-critical moves while retaining enough stroke for full-wafer patterning.

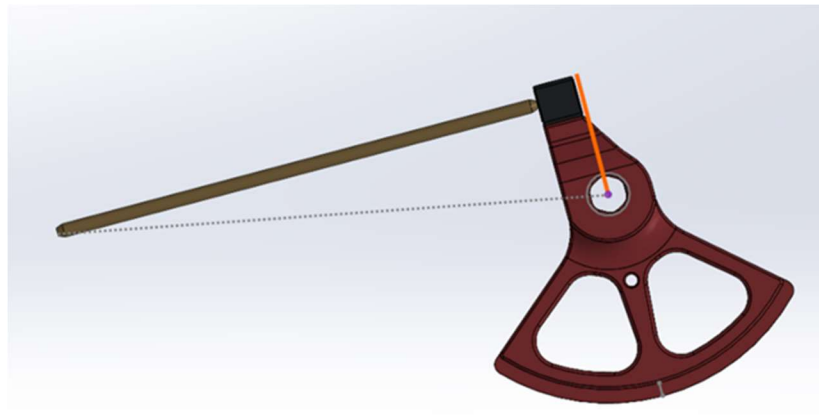


Figure 15: Motor horn attached to rod via ball joint

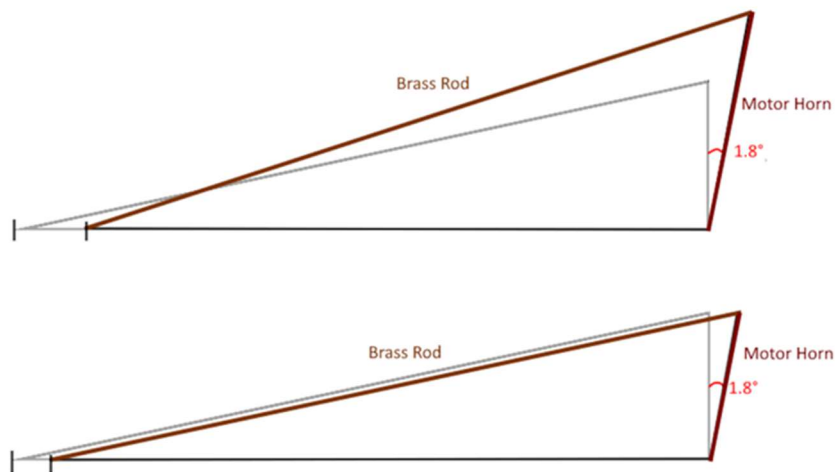


Figure 16: Geometric representation of motor horn & rod

The mechanical design can be simplified and modeled as a triangle where the brass rod and the distance from center axis of motor horn to rod are fixed lengths.

There is a geometric trade-off where the resolution is the density of steps in one length. The stride length of the brass rod can be adjusted depending on how close the rod is to the center of the motor horn. The closer it is the less area it covers; however, it will translate into more steps per length resulting in higher resolution. If the rod was further away from the center of the motor horn, then it can cover more area at the expense of resolution. These trade-offs must be considered depending on the size of the wafer and design complexity per lithography run.

Improved Motor Horn Design (Proposed)

A proposed revision uses a significantly longer motor horn so a larger region of the wafer can be reached without changing the core actuator architecture. The new geometry includes multiple weight-relief holes along the horn body to keep inertia lower despite the increased length.

To reduce compliance, this part is intended to be CNC machined instead of 3D printed. A stiffer machined horn is expected to reduce elastic deflection under load, which helps preserve positioning accuracy during direction changes and higher-speed moves.

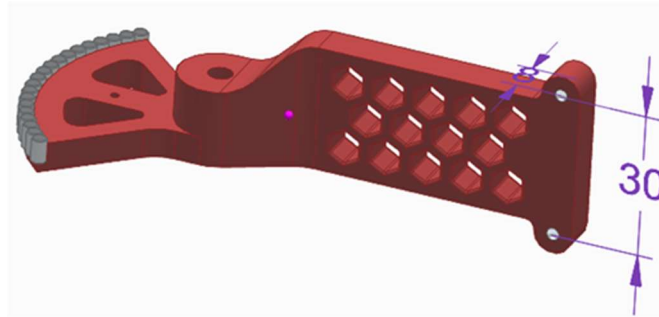


Figure 17: Extended motor horn length

Proposed long motor-horn design with weight-relief holes for larger wafer reach and lower mass; preferred manufacturing path is CNC machining to reduce compliance.

For long continuous sessions, closed-loop feedback is used to prevent small tracking errors from accumulating into laser-path offsets. In magnetic coupling systems, Di Natali et al. report that load-side closed-loop control can achieve smaller steady-state errors, lower ripple, and higher transferable torque before pole-slip than motor-side feedback alone .

Optical System Mount

The optical mount is designed specifically to attach and screw down onto the rails of the T-slot aluminum. The design is modular, making switching out pieces quick as this is still a prototype design, and the optical system can undergo different configurations depending on what is being tested hence the need for something easily adjustable to mount the laser to the frame of the enclosure.

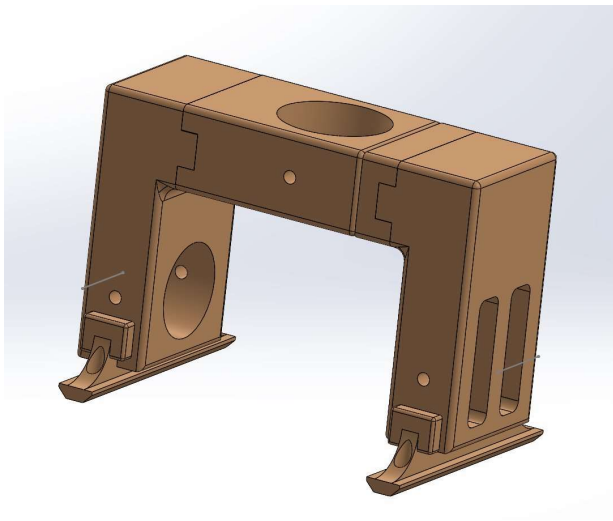


Figure 18: Modular Optic Mount

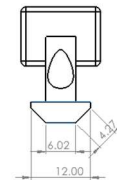
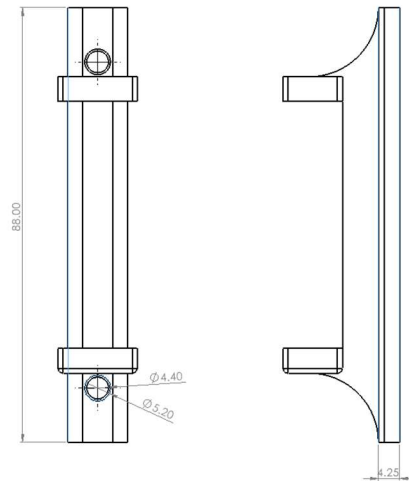


Figure 19: T-slot 2020 slider mount

Alternative XY Stage Design

A modified version of the XYZ stage design is currently being developed, removing the Z stage from the architecture. The Z stage will instead be modified to be independent from X and Y as the precision from the Z stage doesn't matter so it will be mechanical screw style motors adjusting the height of the platform that carries the XY stage.

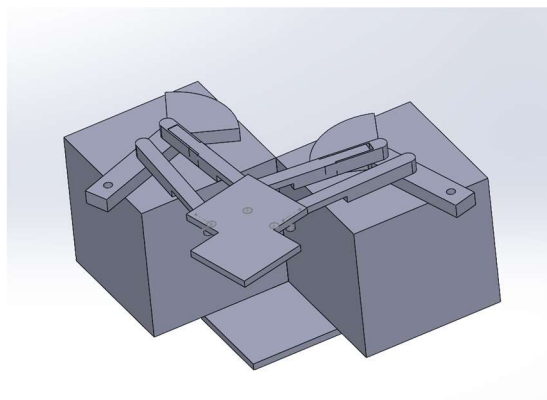


Figure 20: XY Stage Concept

Performance Targets

Stage performance requirements

Metric	Target
Resolution	100–500 nm
Accuracy	300–750 nm
Repeatability	300–750 nm

Laser Field Placement and Theta-Stage Strategy

Initially, we planned to keep the laser-writing field centered on the wafer platform. In practice, the currently achievable XYZ coverage is limited (about 20 mm × 20 mm), and even with the expanded design under development, preserving current resolution while scaling travel is non-trivial. In a conservative scenario, available XY travel may only reach about 30 mm × 30 mm.

With a center-only approach, this limited field reduces usable wafer coverage even if theta rotation exists. We therefore propose combining an off-center writing field with a theta stage so the effective writing region can be rotated to different wafer quadrants. The lithography pattern can then be software-rotated to match the new orientation, allowing continued use of a small high-resolution XY field over more of a 4-inch wafer.

If the theta stage is motorized, this approach also enables more automation because users would not need to manually reposition to each quadrant between writes.

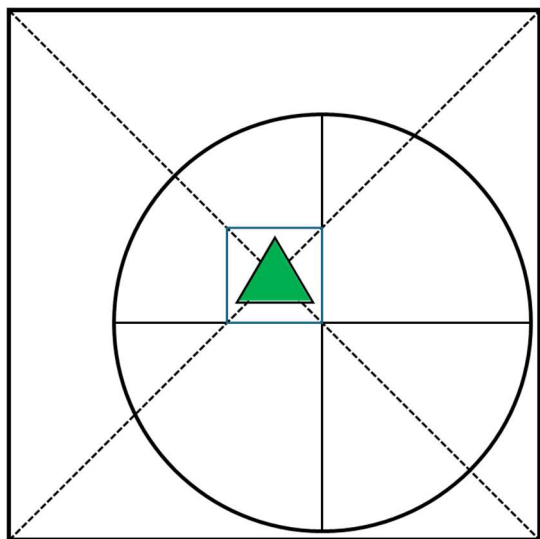


Figure 21: Centered laser-writing field concept, where limited XY travel restricts the wafer area reachable per setup.

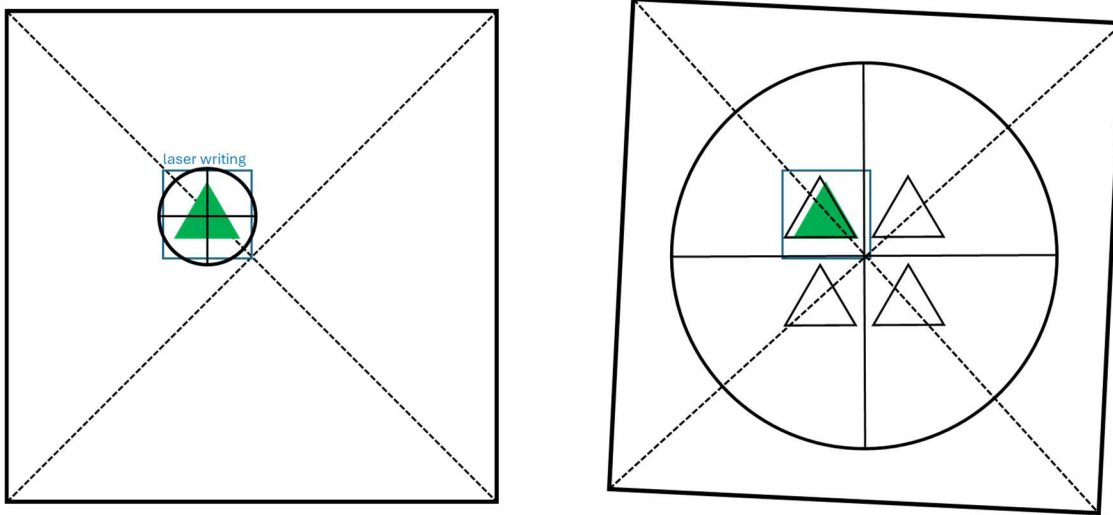


Figure 22: Theta-stage concept: rotate the wafer-relative writing field to target a different quadrant, then apply software-defined pattern rotation for continued high-resolution writing.

This strategy supports (M5.1: stage accuracy, M5.2: improved precision, S1: GUI interface for rotation-aware workflow).

Theta Rotation/ Stage

This strategy supports (M1: Theta stage)

We will integrate a manual theta stage to satisfy this requirement. Pictured below (Figure 22.1).

R Axis **40×40**



Figure 22.1: Manual Theta Stage [11]

Microscope Turret

This specification supports (M6: Microscope Turret to support multiple lens)

We will include a microscope turret because we need to be able to get different dot sizes from the laser.



Figure 22.2 Microscope turret [13]

Continuous-Use Risks and Next Prototype Improvements

Potential long-duration error sources include motor heating, external vibration, and thermal expansion in metal transmission components. NEMA17 motors commonly reach about 60 °C, while PETG fixtures can begin softening near 70 °C–80 °C, reducing stiffness and increasing alignment drift.

The next prototype will implement the following improvements:

- Replace PETG motor mounts with CNC aluminum mounts to increase stiffness, reduce tolerance drift, and improve heat dissipation.
- Use carbon-fiber rods to reduce thermal expansion over long runs.
- Move rods closer to the rotation axis to increase positioning resolution, with an explicit trade-off of reduced workspace.
- Replace temporary wiring with JST connectors to improve electrical reliability under vibration and repeated operation.

XYZ Stage Construction Instructions

This section describes the construction procedure for the stage.

Step 1: Mechanical Frame

The base frame is constructed from T-slot aluminum extrusions.

This step fulfills (M0.3: benchtop compatibility, M1.2: thermal resistance).

Assembly steps:

1. cut aluminum extrusion segments
2. assemble rectangular base frame
3. mount vibration isolation feet

Step 2: Motor Mounts

Motor mounts are CNC aluminum components.

This step fulfills (M1.1: rigid motor mounting, M1.3: vibration isolation).

Mounts are attached using M5 bolts.

Step 3: Motion Linkage

Components:

- carbon fiber rods
- ball joints
- preloaded tension string

This design provides sub-micron positioning resolution.

Camera Alignment System

The camera alignment system implements requirement M4.6 (camera sees laser path).

The camera views the wafer through the same objective used for laser writing.

A dichroic mirror reflects visible light to the camera while transmitting UV laser light.

This coaxial configuration ensures that the camera observes the same focal point as the laser.

Dual Ring-Aligner Fine Calibration

To align the laser spot with the red GUI crosshair at sub-micron scale, two ring aligners are required in the optical mount. Because the focused laser spot can be as small as $0.6\ \mu\text{m}$ in diameter, very small optical misalignment can produce a large registration error during writing.

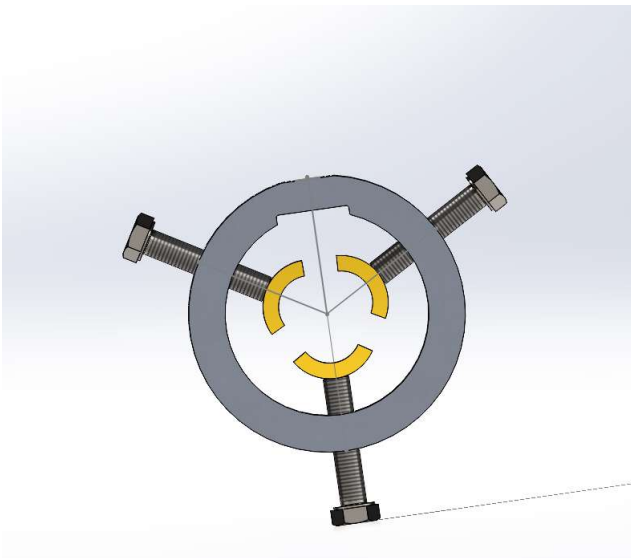


Figure 23: Ring-aligner mechanism used for fine optical correction. Using two ring aligners provides sufficient coupled adjustment freedom to center the laser spot on the GUI crosshair. We are using M5 screws providing 0.8mm per turn which could provide a laser dot movement resolution of $2.22\ \mu\text{m}$.

Calibration is performed with room lighting minimized while observing through the objective lens. The operator captures camera images of the laser spot in a dark condition, compares spot position against the red GUI crosshair, then iteratively adjusts both ring aligners and

rechecks alignment until the spot is optimally centered. This trial-and-error workflow is used before exposure and supports (M4.6: camera sees laser path, S4.5: camera video to GUI).

These ring-aligner could be easily 3D printed or machined on metal, can easily attached to our T-slot aluminum enclosure, and will have a locking nut between the outside of the grey collar and the bolt to prevent long term drifting from what is set. The goal is to make sure the end user doesn't have to recenter it themselves to satisfy ease of use.

Software Architecture

The software system fulfills (S0: GDS file ingestion, S1: GUI interface, S4: communication with microcontroller).

Processing Pipeline

1. import .GDS design file
2. convert geometry to internal representation
3. perform lithography simulation
4. generate G-code instructions
5. transmit commands to microcontroller

Layer-to-Layer Alignment in GUI

The GUI is required to align each lithography layer on top of previously exposed layers using wafer alignment marks. During alignment, the live camera image is shown in the same coordinate context as the planned pattern so the operator can visually match fiducials before exposure.

Because the camera is coaxial with the writing optics, the camera crosshair indicates where the laser is pointing. The user jogs the stage in X/Y (and focus in Z), centers the alignment mark under the crosshair, and then applies the measured offset to the next layer before starting exposure. This reduces layer-to-layer registration error and makes multilayer fabrication practical in normal operation.

This alignment workflow fulfills (S1: GUI interface, M4.6: camera sees laser path, S4.5: camera video to GUI).

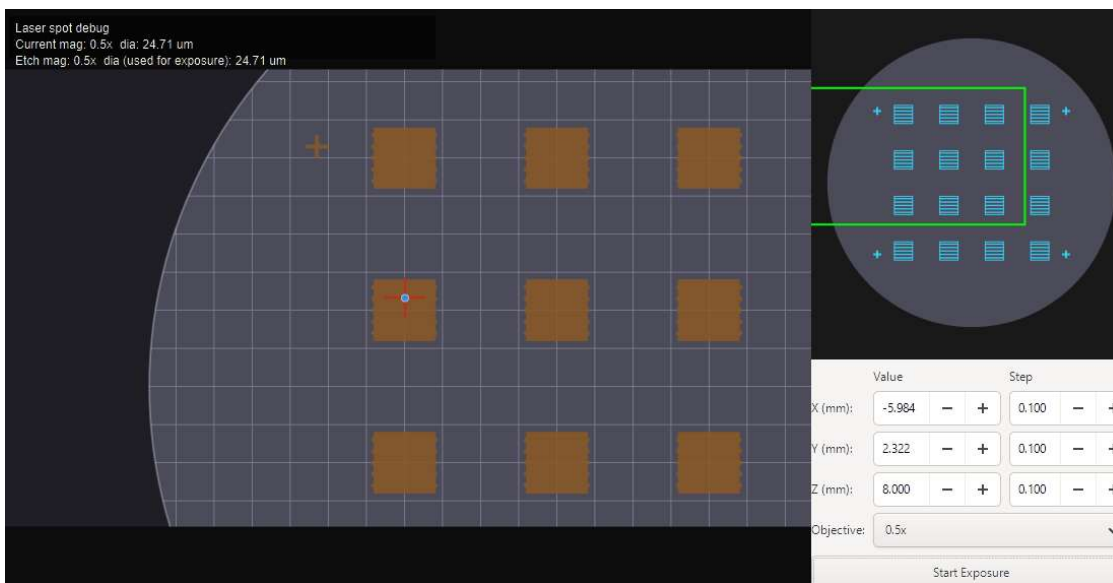


Figure 24: Software GUI view used for multilayer alignment with fiducial marks, live camera feedback, and stage jog controls. The crosshair corresponds to the laser pointing location for layer registration (Responsible: Justin — Software/GUI integration & laser writing algorithm).

A standard wafer alignment-mark arrangement and the resulting overlay error when the next layer is not perfectly registered are shown in the following diagrams based on the WNF Mark Design reference . The misaligned case shows that translation-only correction is not always sufficient: accurate X/Y control is required to remove offset, and theta (rotation) correction is also highly useful to remove angular error and speed up user alignment.

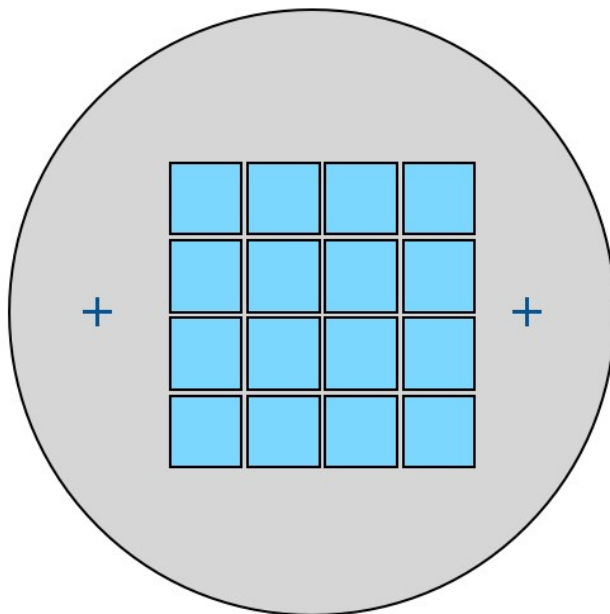


Figure 25: Typical wafer alignment-mark layout used as the target registration geometry for multilayer exposure (adapted from WNF Mark Design reference).

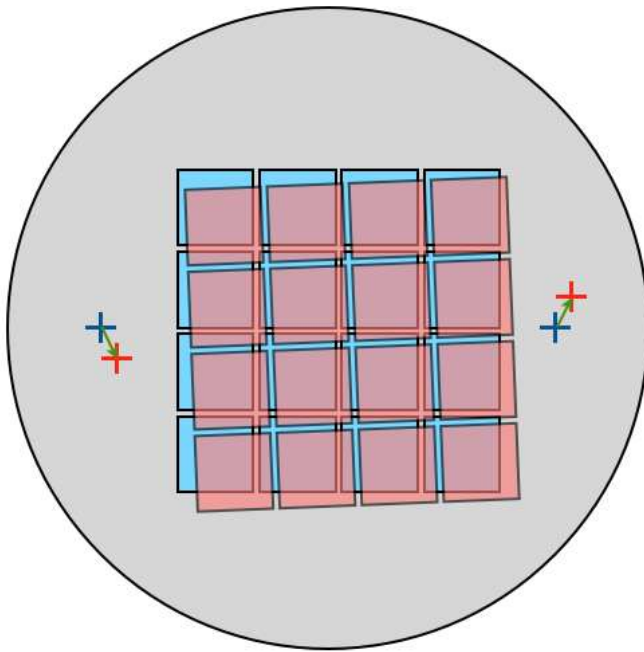


Figure 26: Example layer-to-layer misalignment showing residual X/Y shift and rotation, motivating both fine XY stage correction and theta (rotation) correction in the GUI workflow.

Lithography Algorithm

The lithography algorithm implements requirement S0.1.

Simplified pseudo code version:

START APPLICATION

Initialize GUI

Initialize stage state

Initialize optics state

Initialize camera/simulation mode

Load default design if available

WHILE application is running

 If user load/export GDS files

 load/export design geometry

 Display design on wafer map

```

    Choose wafer size
If user changes X/Y/Z position
    Validate limits
    Update stage position
    Refresh camera and wafer view
If user changes step size
    Update motion increment values
If user changes objective
    Update magnification, NA, focus settings
    Refresh simulation view
If user changes laser power or frequency
    Update exposure parameter
    Send signal to the firmware for different duty cycle
If user changes camera settings
    Apply shutter time, gain, pixel format
If user clicks wafer map
    Request stage move to clicked position
If user starts exposure
    Check and choose appropriate algorithm:
    If polygons exist
        polygon outline +fill
    Else if path exist
        Path only
    Else
        Raster Hit Test
    WHILE exposure is active
        on each timer tick:
            update motion/exposure state
            Sample geometry
            Record exposed points
            Refresh display with drawn from recorded points
    END WHILE

```

If camera mode is active

Show live image

Else

Show simulated/wafer view

END WHILE

EXIT APPLICATION

Three path generation modes exist:

- PolygonOutlineFill:
 - For closed polygons, the algorithm first performs an outline stage and then a fill stage. During the outline stage, each polygon edge is traversed segment by segment using a small laser spot size to improve boundary definition. Three outline passes are executed: one on the nominal polygon edge, one offset slightly outward along the segment normal, and one offset slightly inward. This produces a thicker and more reliable boundary trace before any area fill is attempted. Once all outline passes are complete, the simulation computes the polygon bounding box and switches to fill mode.
 - In the fill stage, the polygon interior is exposed using a serpentine raster strategy. The algorithm starts near one corner of the polygon bounding box and scans horizontally across the shape, row by row, alternating direction on each successive row. At each step, it evaluates whether the sampled position lies inside the polygon using a point-in-polygon test. Exposure is applied only for samples that are inside the polygon, while samples outside the shape are treated as laser-off motion. To improve boundary accuracy, the current implementation subdivides each horizontal move geometrically into smaller internal samples for laser gating decisions but only issues a single actual stage motion command to the path planner for the final endpoint of that tick. This preserves polygon-edge accuracy without flooding the motion system with excessive commands.
- PathOnly
 - The algorithm uses a simpler tracing mode. Each path is treated as a sequence of connected line segments. The stage first moves to the initial point of the path with the laser off to avoid unintended bridging and then advances along each segment incrementally according to the computed exposure step size. Laser firing occurs while traversing the intended path, so the result is effectively a stroked exposure along the imported line of geometry rather than an area of fill. When one path is complete, the algorithm proceeds to the next until all paths have been processed.
- RasterHitTest
 - The algorithm is the fallback exposure algorithm used when the imported design does not provide polygon or path data suitable for the dedicated

PolygonOutlineFill or PathOnly workflows. In this mode, the program does not explicitly trace polygon edges or path segments. Instead, it performs a bounding-box scan over the design area and determines whether each sampled stage position lies on exposed geometry by calling the design's hit-test function. This makes the method more generic, since it can operate directly on the rendered design representation without requiring a separate polygon fill or path traversal stage.

Additional Features (Important for the main purpose of user friendly):

- An additional design feature is quadrant-restricted exposure for larger wafers. For wafer sizes above 1 inch, the program can constrain exposure to one of the quadrant regions and rotate the scene between passes so that all geometry is eventually exposed while maintaining a fixed processing region. For 1-inch wafers, this restriction is disabled, so the full design can be exposed directly without rotation.
- The position controls allow direct manual adjustment of the X, Y, and Z stage coordinates, while the step controls define how much each axis changes during incremental movement. X and Y are limited to the valid travel range, and Z is clamped by a safety minimum. Together, these controls support both coarse positioning and fine alignment.
- The objective selector changes the active optical preset, which affects magnification, numerical aperture, focus position, field of view, and laser spot size.
- The camera settings include shutter time, gain, and pixel format, which are applied to the live camera feed when camera mode is active. The application can switch between simulation mode, where the wafer and exposure process are rendered, and camera mode, where the live image is shown. Grab controls allow single capture, continuous acquisition, or stopping the stream.
- The camera settings include shutter time, gain, and pixel format, which are applied to the live camera feed when camera mode is active. The application can switch between simulation mode, where the wafer and exposure process are rendered, and camera mode, where the live image is shown. Grab controls allow single capture, continuous acquisition, or stopping the stream.
- The wafer map and GDS features provide the design workflow. GDS files can be opened, displayed on the wafer map, and used as the source geometry for exposure planning. The wafer map also shows the current camera footprint and can be used for click-based movement requests. Additional features such as autofocus, alignment overlays, scale bars, and the start/stop exposure button make the GUI a combined tool for positioning, imaging, and lithography simulation.
- Laser power and frequency sliding bars provide the main exposure settings, with values updated interactively in the interface. These parameters feed into the exposure model and influence how the simulated lithography process behaves. The laser power is controlled through pulse-width modulation, by changing the duty cycle which reflectively changing the average value of the pulse wave-width

$$D = \frac{t_{on}}{T}$$

$$P_{laser,avg} \approx D \cdot P_{laser,peak}$$

D : duty cycle $\in ($

t_{on} : on – time (s)

T : Laser period(s)

$P_{laser,avg}$: Average laser power

$P_{laser,peak}$: Maximum laser power

Experimental Design Showcases

The following four examples show the current experimental design flow for the laser writing algorithm: (1) a simple rectangle defined in a .GDS layout, (2) the planned infill writing

strategy, (3) a successful simulated write result, (4) a simulation case that intentionally includes mechanical limitations, and (5) a simulation that demonstrate the capability of performing different shapes.

These showcases support (S0.1: lithography path generation), and they are also used to evaluate motion-quality effects tied to (M5.1: stage accuracy) and (M5.2: improved precision).

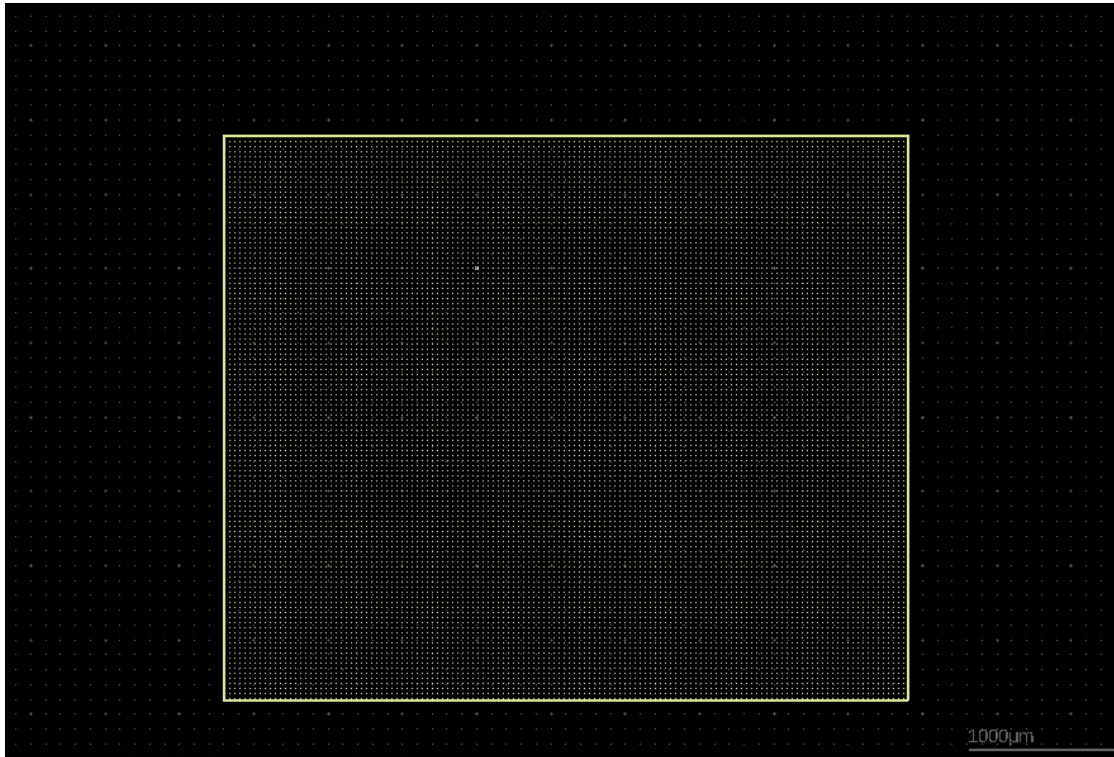


Figure 27: Experimental showcase step 1: .GDS design of a simple rectangle used as the input geometry for algorithm evaluation (Responsible: Justin — Software/GUI integration & laser writing algorithm).

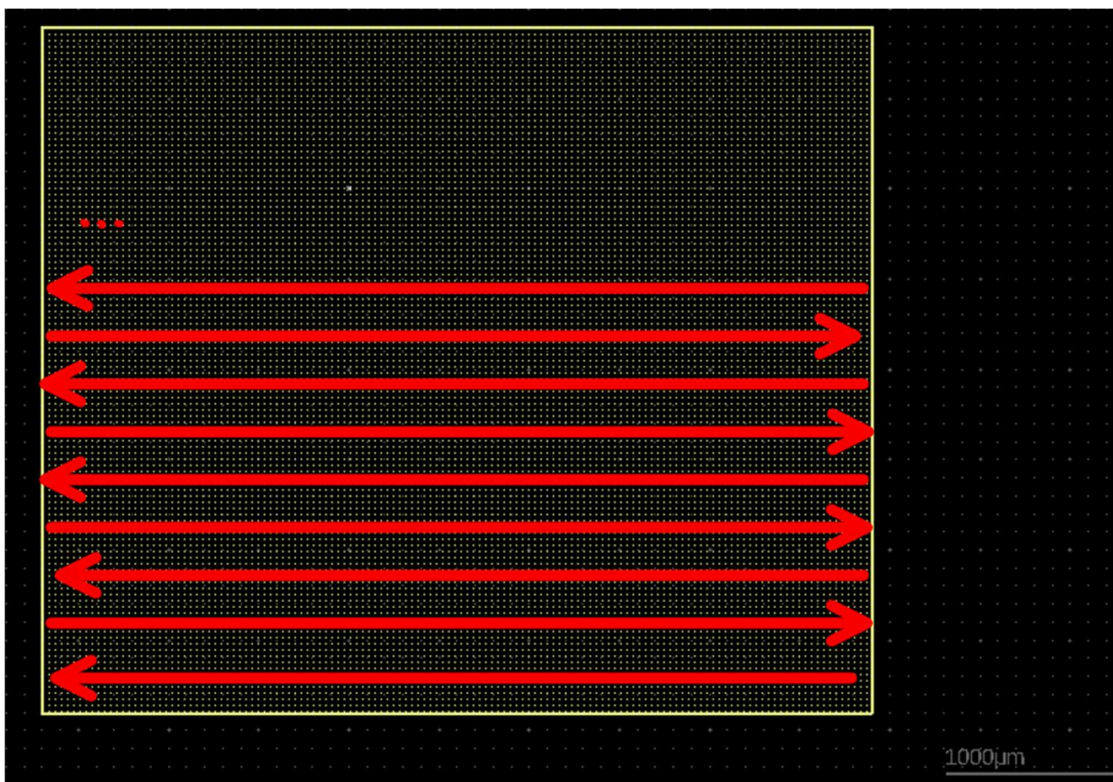


Figure 28: Experimental showcase step 2: planned serpentine infill strategy for exposing the rectangle area.

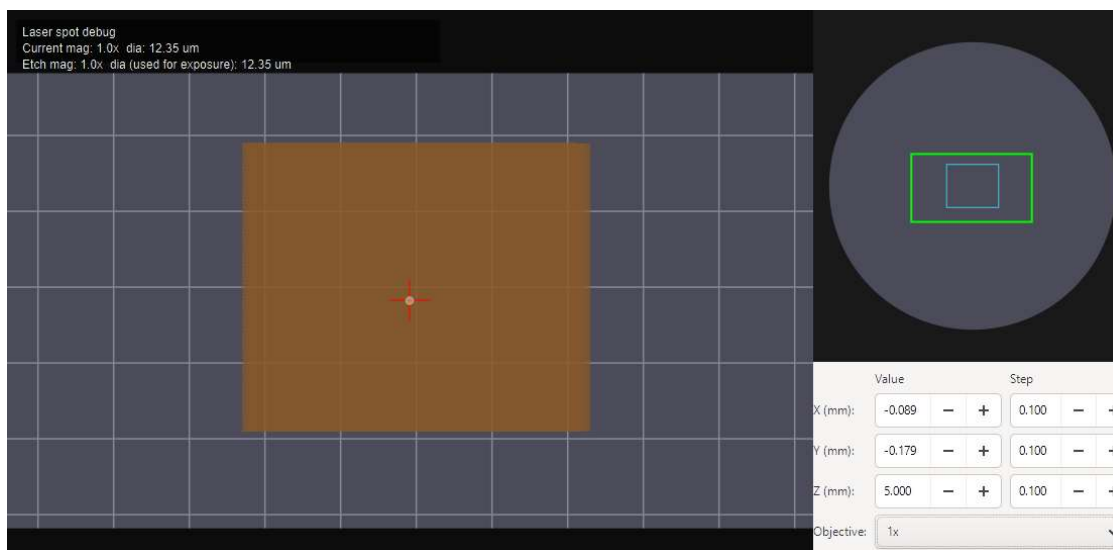


Figure 29: Experimental showcase step 3: successful simulated result for the infilled rectangle write.

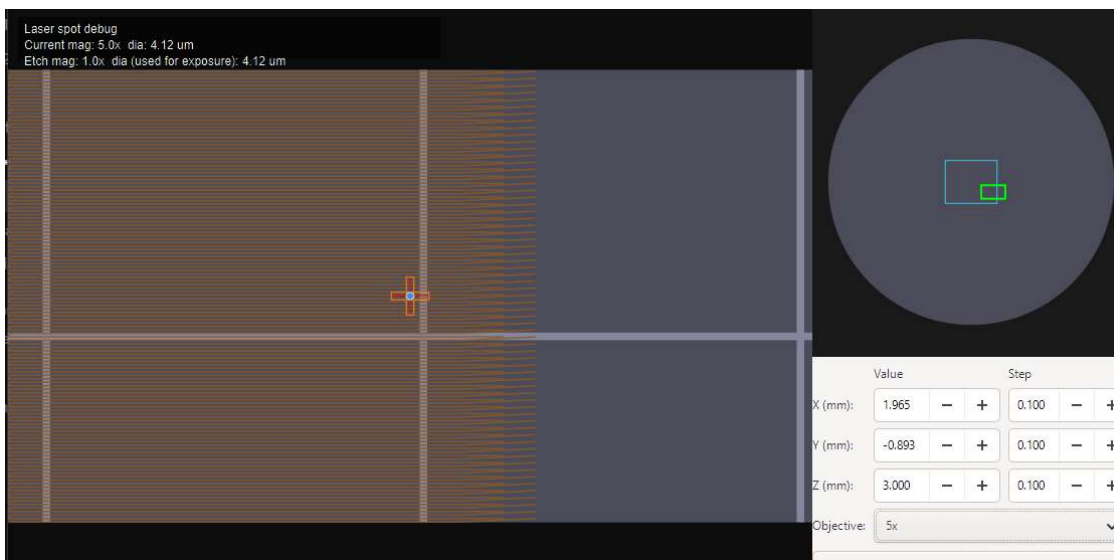


Figure 30: Experimental showcase step 4: simulation with real mechanical limitations enabled. Current behavior reflects motion constraints, and the next step is to also simulate system-accuracy limitations explicitly.

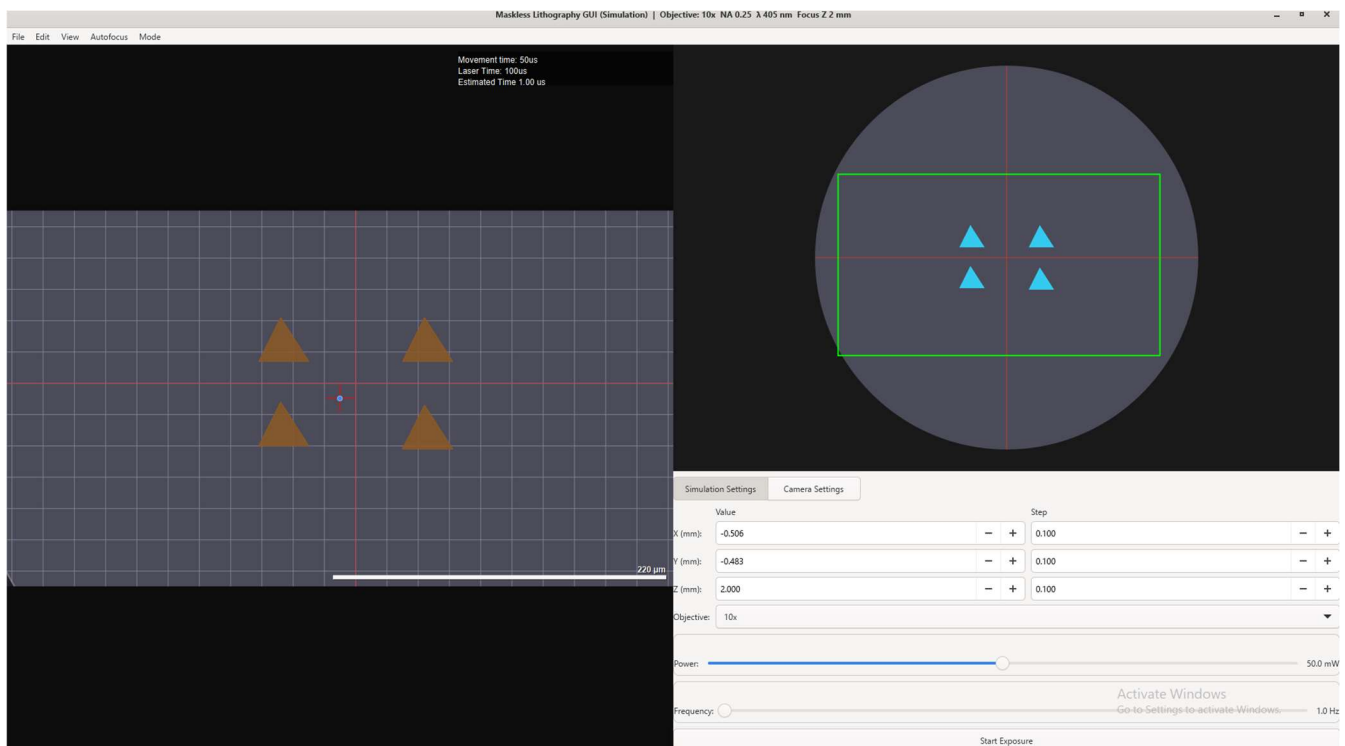


Figure 31: Experimental showcase step 5: simulation for triangle shape, indicate the capability of performing different shapes with diagonal components to them

Zig-Zag Path Case Study and Algorithm Improvement

For .GDS import and rendering, we use the C++ GDStk library, and for the interactive operator interface we use the C++ GTK 4 library. This zig-zag test case is intentionally more difficult than the rectangle case and is used to stress path parsing and execution logic for non-orthogonal geometries.

The original approach treated the shape primarily as polygon infill and scanned left-to-right across the geometry, firing when parts of the path came into view of the laser dot. This produced a slow and poor laser writing strategy because many sweeps do not follow the intended path entity directly, making the XYZ stage move through a lot of empty space.

$$L = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Using line entities and segment-length computation reduces unnecessary traversal and gives a cleaner, more efficient write path for this geometry.

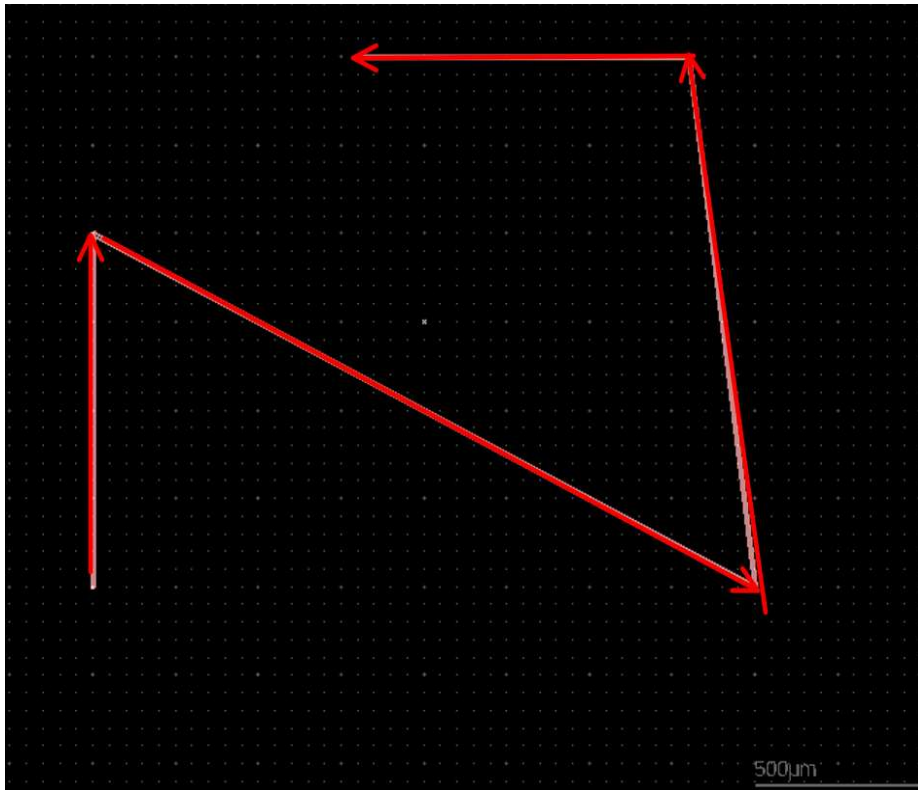


Figure 34: Improved PathOnly algorithm: line-entity-based traversal for the zig-zag geometry, yielding faster and more efficient path generation than polygon-only infill.

This case study supports (S0: GDS file ingestion, S0.1: lithography path generation, S1: GUI interface).

This improvement can help us with making more complex shapes like triangles. Currently we face issues with infilling large shapes as the laser dot (of around $\sim 1\mu\text{m}$) is too small to infill shapes, so we're looking into a 0.05 NA objective lens that could produce a far bigger laser dot ($\sim 50\mu\text{m}$) used specifically for infilling shapes. This necessitates the need for an objective lens turret that could allow the software to automatically switch lenses based on whether or not the outline of shapes is being made or if we are infilling a shape.

Environmental Enclosure Filtration System

Air Filtration

The air filtration subsystem implements requirement M7.1 (contamination control).

The air filtration system is engineered to maintain a Class 5 cleanroom environment within the enclosure, a critical requirement for achieving a 0.7um feature size target without particulate interference. To ensure the air stream remains unobstructed by the optical column and robotic assembly, we have implemented a Horizontal Laminar Flow configuration. This design delivers purified air directly across the wafer surface at a target horizontal velocity of 0.45 m/s (approximately 88 fpm), which provides the necessary kinetic energy to sweep particles away from the sensitive exposure area.

Modeling and Volumetric Requirements

To achieve this precise environment, our modeling focuses on the air delivery opening directly at the wafer, which measures 204 cm² (approximately 0.22 sq ft). Using the fluid mechanics principle $Q = v * A$, we calculate the required volumetric flow rate at the wafer surface as follows:

$$Q_{target} = 88 \text{ fpm} \times 0.22 \text{ sq ft} \approx 20 \text{ CFM}$$

However, the internal design must account for significant static pressure losses caused by the HEPA/ULPA filter media and the intake pre-filter. We predict a 90% loss in effective flow from the blower's "free air" rating due to these resistances. Consequently, to maintain the required 20 CFM at the wafer, the system necessitates a blower with a much higher capacity:

$$Q_{required} = \frac{20 \text{ CFM}}{0.10} = 200 \text{ CFM}$$

Based on these predictions, we have selected a 4-inch air blower with a free-air rating of 204 CFM. The inclusion of speed control allows for precise calibration, ensuring we can meet the 0.45 m/s goal even as filters accumulate particulates over time.

Alternative Design and Vibration Mitigation

Recognizing that the 204 CFM blower may introduce mechanical vibrations that could blur sub-micron lithography, we have developed a backup design to ensure system stability. This alternative involves housing the blower and coarse pre-filter in a separate external enclosure, physically decoupled from the main device. In this configuration, air is delivered to the internal 204 cm² plenum via a flexible air duct. This satisfies the requirement for a distinct and reasonable design alternative while addressing the critical risk of motor-induced oscillations which could cause vibrations within the view of the laser and the camera.

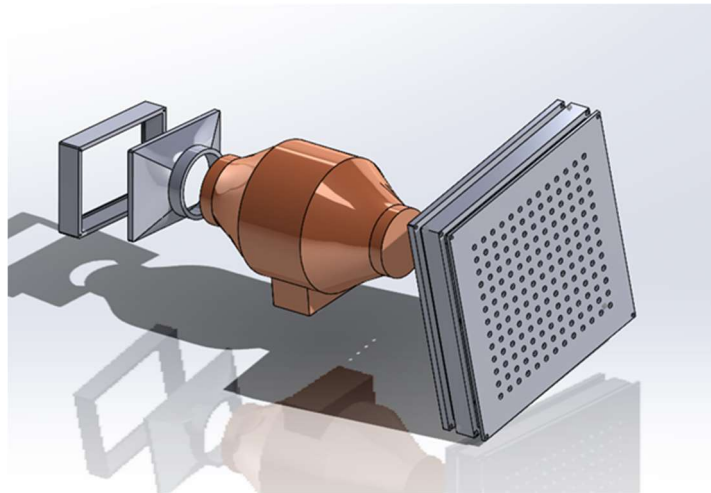


Figure 35: Air filtration assembly concept showing the 4-inch blower with speed control, 3D-printed vents, and potential aluminum tube interface (Responsible: Tony — environmental filtration enclosure).

Enclosure Design

The enclosure is designed as a high-precision housing that fulfills both the structural requirements of a benchtop lithography system and the rigorous safety standards for a Class

1 laser device. The core structural frame utilizes aluminum extrusions, selected for their modularity and the flexibility they provide when mounting internal optical brackets and sub-assemblies. All internal mounting components are fabricated via 3D printing with PETG for rapid prototyping of complex geometries, or CNC machining for high-load sections requiring superior mechanical rigidity.

Safety Mechanisms and Laser Containment

To maintain Class 1 safety status, the enclosure is a fully light-tight, closed system designed to prevent any accidental user exposure to the laser source.

- **Loading Interface:** Substrates are introduced via a CD drawer-style mechanism featuring a spring-loaded lock and an integrated position sensor to ensure the drawer is fully retracted and sealed before exposure begins.

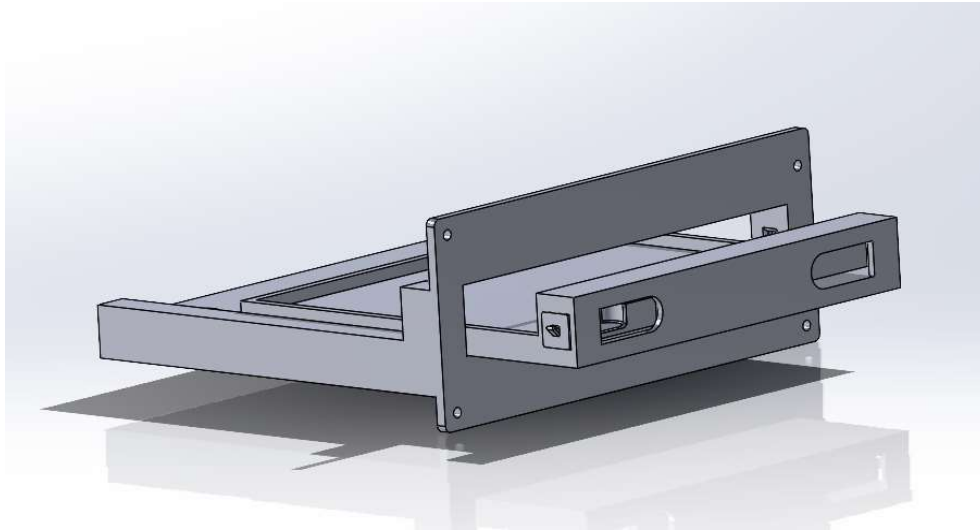


Figure 36: *Flap-door wafer-loading module for 4-inch wafer-case insertion with internal glove operation, enabling laser-safe and contamination-controlled loading*

- **Interlock System:** The aluminum/polycarbonate panels are equipped with mechanical limit switches that detect if any part of the housing is removed, triggering an immediate hardware-level laser cutoff.
- **Emergency Controls:** A prominent, physical Emergency Stop (E-Stop) button is mounted on the exterior for immediate system de-activation in the event of a malfunction.
- We do plan an LCD display or LED indicator towards Capstone B that will indicate that the laser is on. An LCD display could have more information for the user.

User Interface and Interaction

User interaction is primarily managed through a dedicated software suite, supplemented by hardware feedback directly on the enclosure.

- **Status Monitoring:** A mini status screen integrated into the front panel provides real-time telemetry, such as current process stage and safety interlock status.
- **UV/Laser safe Transparent polycarbonate panels:** allow the operator to see the process and interpret if something goes wrong.

Modeling and Alternatives

The enclosure's dimensions are modeled to provide a stable center of mass, preventing vibrations from the air filtration system from coupling with the optical breadboard.

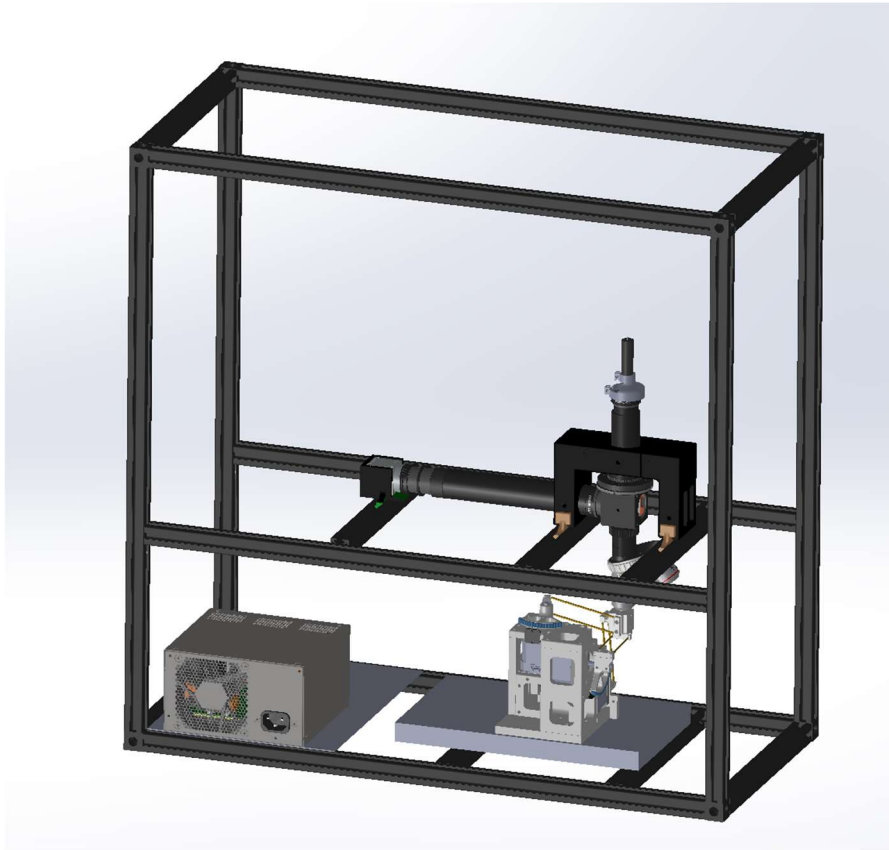


Figure 37: Enclosure without panels, screens and e-stop. Aluminum extrusions were used for the mounting flexibility of the interior components.

Laser Safety System

The system satisfies requirement G1.1 (laser containment) and G1.2 (interlocked laser shutdown).

Safety components:

- light-tight enclosure
- interlock switch
- emergency stop
- automatic laser shutdown

Salvaged Safety Hardware for Armin's Electrical System

For the electrical safety chain, Armin identified reusable hardware from salvaged parts: a key-reset emergency STOP button with a switch block mounted at the rear, and a set of scavenged breakers that can be repurposed for branch isolation in the interlock path.



Figure 38: Salvaged emergency STOP button (front view) selected for integration into the fail-safe laser shutdown chain (Responsible: Armin — power, electrical interlock, and safety design).

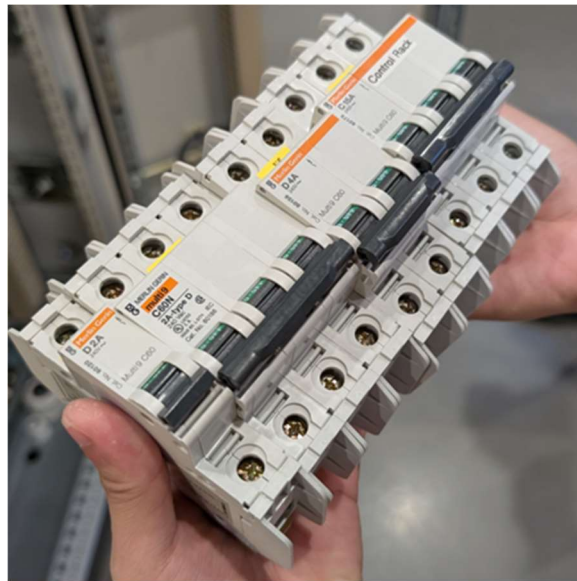


Figure 39: Scavenged circuit breakers are considered for use in the breaker/interlock safety path to isolate laser and stage power during fault or emergency events (Responsible: Armin — power, electrical interlock, and safety design).

Interlock Circuit Designs

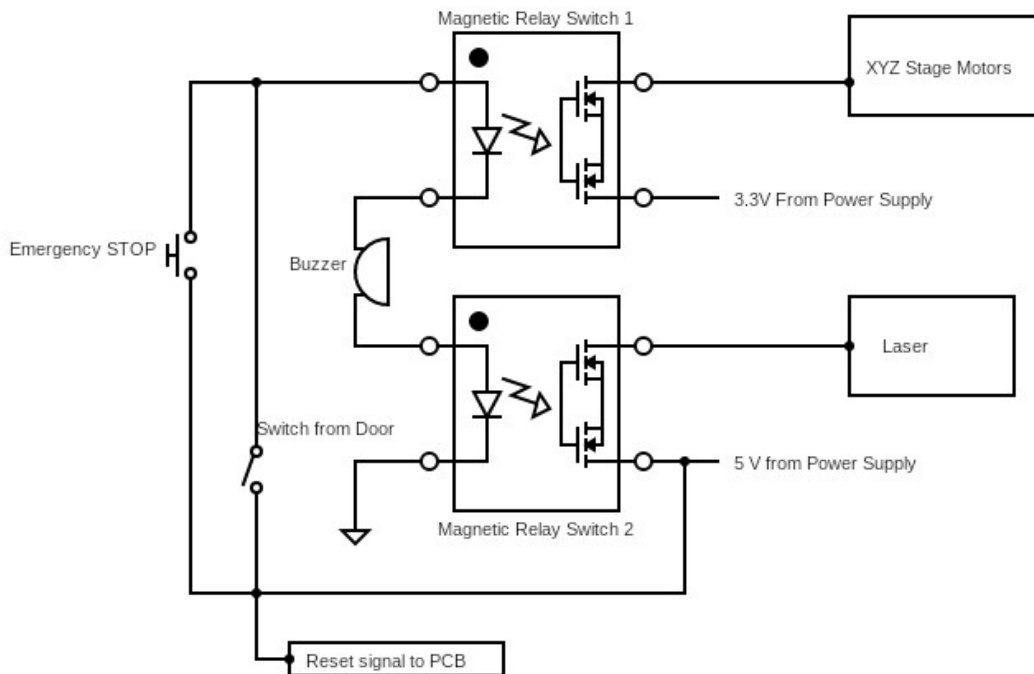


Figure 40: Breaker and interlock safety architecture for laser power isolation and enclosure interlock control (Responsible: Armin — power, electrical interlock, and safety design).

Pictured above is the Emergency Stop Circuit, which is designed to cut power to both the Laser, as well as the XYZ stage under two conditions: when the emergency stop button is activated, or when the enclosure door is opened. This dual function is achieved by placing the emergency door switch in parallel with the emergency stop button. The Laser and XYZ stages were chosen as the only two components for having power cut off in an emergency stop event. The laser due to obvious hazards should it lose containment, and the XYZ stage was selected to instantly freeze the lithography process in place. The other systems are to remain active, allowing for user control to reset the device, or to restart the lithography process.

In the design above, the relay switches are normally closed, and when power is diverted by the emergency switches, the relays will be triggered as their internal coils and diodes are powered. This way the power is cut off only from the laser and XYZ stage but still left intact for other systems of the device to remain functional. The relays selected for the design are a 5V relay with optocoupler model DK B07957459Q, which was chosen for its powered design, as when the relay is tripped, it can be set so that it will not reopen until power is reset. This functionality will be implemented by sending the reset signal back into the relays, as well as to the PCB. The buzzer in this circuit is a temporary design choice to allow for quick testing and verification that the circuit works correctly, and will be removed at a later stage.

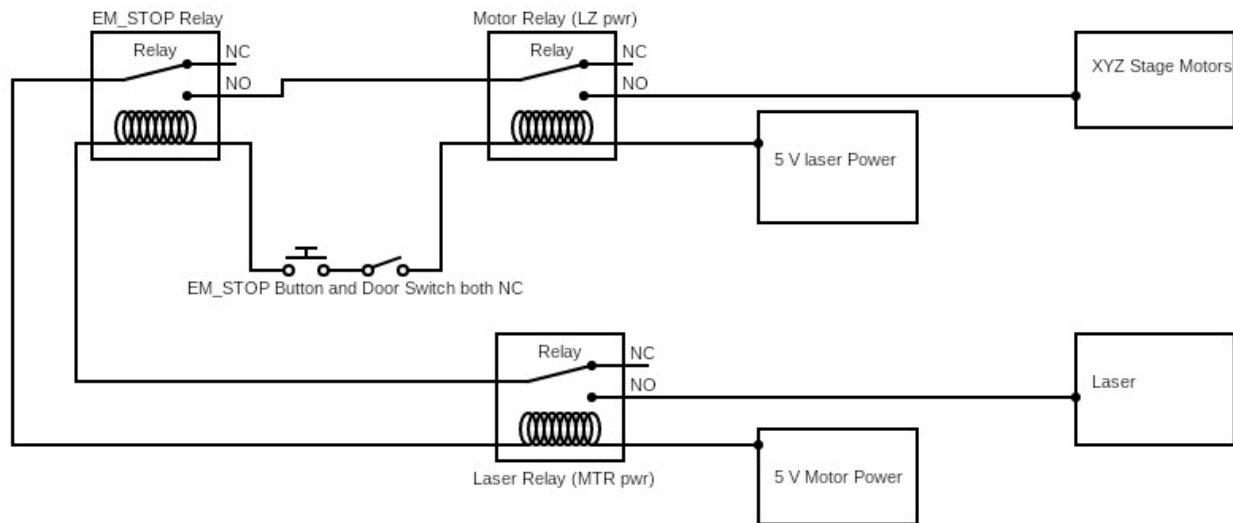


Figure 41: Breaker and interlock safety architecture for laser power isolation and enclosure interlock control Backup design

Above is an alternative design for the emergency stop circuit. This design is arranged in a “Normally Closed” (NC) configuration. This means that the circuit power is always flowing through the emergency stop switches, and the circuit is broken when the switches are engaged. This circuit arrangement is generally considered to be safer for dangerous machinery, as should any component fail; the device will be rendered inoperable. Should the emergency switch fail, the device would also be inoperable, as the circuit will be open.

This design functions by independently powering the relays, thereby keeping them closed, and then allowing the circuit to power the mechanical components of the XYZ stage and the laser. Should power be interrupted, the relays will automatically open the circuit, and should the emergency switches be engaged, the relay power will also be lost, opening the circuit and shutting down the laser. To achieve this, this circuit has placed the EM stop button and the Door Switch in series, oppositely configured to the previous design. Should our previous design in figure 40 have any unforeseen flaws, or should it not perform as safely as expected, this second configuration will be implemented instead. As the laser is a dangerous component, and as one of our major goals is to achieve a class 1 device prototype, we deemed it necessary to have at least two designs ready for the emergency stop circuit to help meet this requirement.

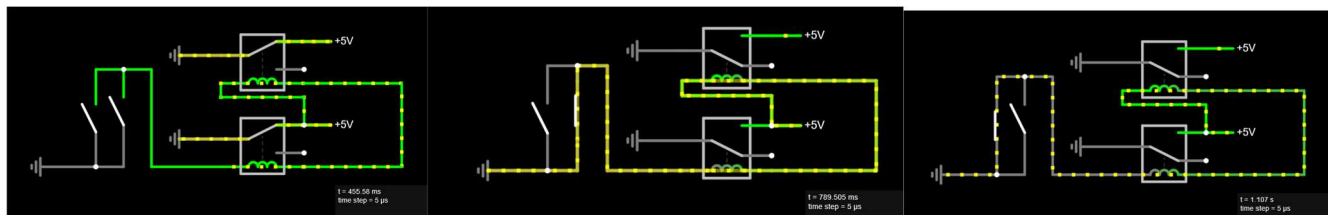


Figure 42: These three Images are a simulation of the first emergency stop circuit (Figure 40). As can be seen, when either of the switches are engaged, the power is diverted and the relays are switched open, powering down the loads in series with the relays. In the left most side, we can see both switches are open and power only flows from the power supply to the load, through the relays.

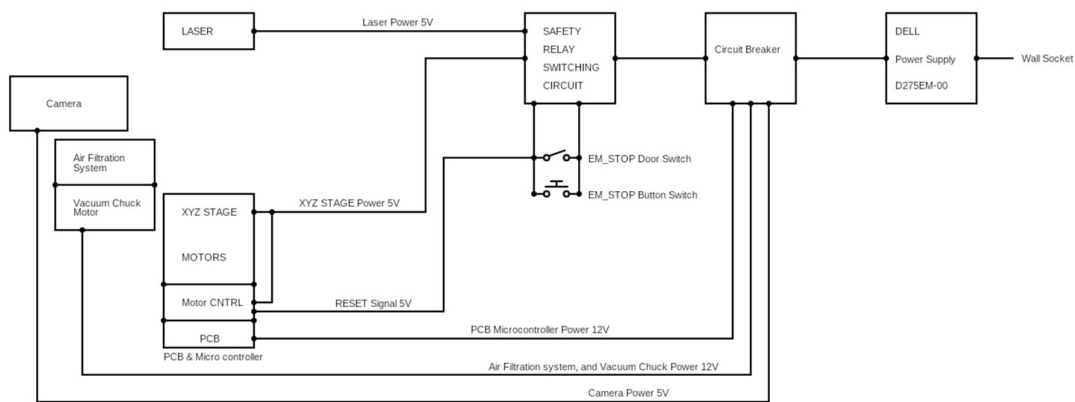


Figure 43: System-level breaker and interlock overview for Armin's electrical safety chain, showing power distribution and shutdown paths between PSU, breaker, relay switching circuit, laser, stage, and support loads.

The architecture of the overall electrical system is as follows. A Dell computer power supply feeds all power levels for each of the different components into a circuit breaker which was salvaged from the previously mentioned decommissioned devices. The power supply used is a Dell model D275EM-00 Variable Switch Power Supply. This was selected for its stable output, even under low current loads, as well as its diverse power output levels which are split into multiple 5V, 3.3V, and 12V channels, each voltage level having at least 4 independent output pins (for pin layout, please refer to figure: 47 in next section). This allows flexibility in powering the system, as well as allowing each subcomponent to be powered independently, allowing for greater modularity in design and repairs of the overall system. The voltage levels of the power output by the power supply are standard levels compatible with most components, therefore choosing this power supply eliminated the needs for additional voltage level circuits, or internal transformers between subcomponents, reducing overall complexity.

This architecture, and the inclusion of the circuit breaker module, was selected for its modularity. As can be seen, the circuit breaker feeds power directly to each component individually, with no components being connected in series with each other. This design choice provides multiple advantages. First, as each component is electrically isolated from all the others, any failure within one subcomponent can be spotted more easily as the failure will be isolated to the component itself. Secondly, the modularity of this design allows for individual testing of each subcomponent without needing to provide power to other systems. Furthermore, the inclusion of the circuit breaker will isolate the power supply and other components from each other. Should a critical failure or short circuit occur, it will remain isolated by the circuit breaker.

This design fulfills (G1; Class 1 laser device within normal operation within it's enclosure).

Enclosure opening and Lab space Laser Safety

A compact flap-door loading module is used to insert 4-inch wafer cases without opening a large enclosure door. The wafer case can then be opened using a glove from inside the enclosure, preserving laser containment and reducing contamination ingress during loading. This design fulfills (G1.1: laser containment, G1.2: interlocked laser shutdown, M7.1: contamination control).

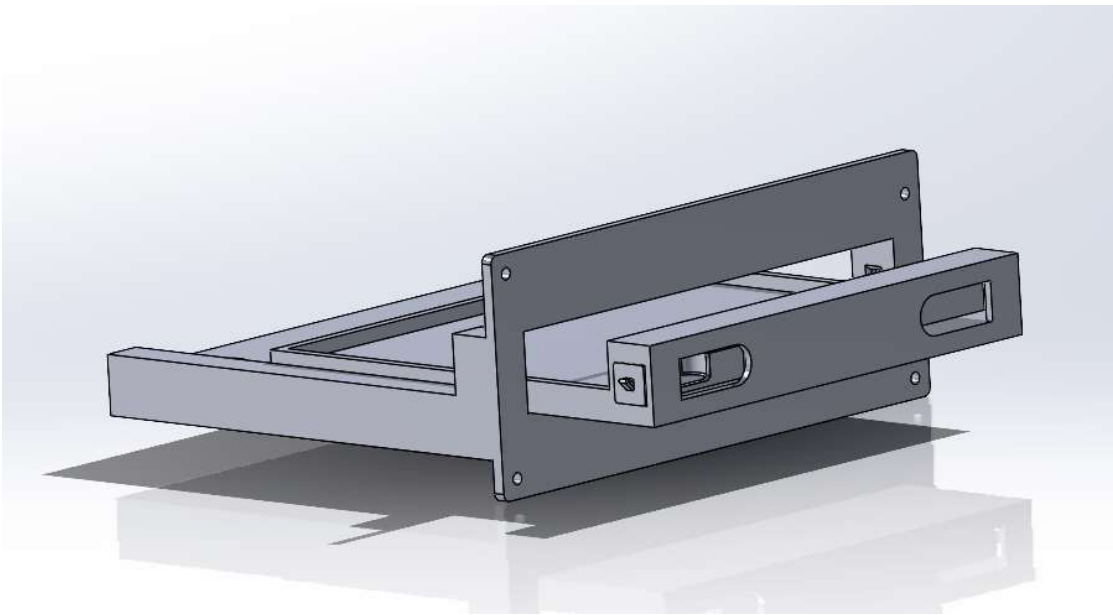


Figure 45: Flap-door wafer-loading module for 4-inch wafer-case insertion with internal glove operation, enabling laser-safe and contamination-controlled loading. A roller switch in the back left side of the drawer rail will trip the safety circuit when the drawer is open.

(Responsible: Armin — power, electrical interlock, and safety design).

The printer-room glass window is additionally covered during laser operation to block line-of-sight stray light, and a caution notice is posted instructing users to knock before entering when printers are in use. This adds an administrative and physical barrier on top of the electrical interlock chain.



Figure 46: Printer-room window cover and caution signage used to reduce stray-laser exposure risk and control entry during operation (Responsible: Armin — power, electrical interlock, and safety design).

In summary, the flap-door interlock and emergency STOP are wired in a fail-safe breaker chain. Under normal operation, both safety relays remain energized and allow power to the laser driver and XYZ motor-drive path.

If the user opens the flap door or presses the emergency STOP button, the relay chain de-energizes immediately, which opens both relay contacts and physically removes power from the laser and stage motors. This is a hard power cut (not only a firmware command), so motion and emission stop even if the controller software is frozen.

A reset signal is then sent to the PCB, and operation can resume only after the door is closed, the E-STOP is released, and the system is intentionally reset by the user.

This design fulfills (G1; Class 1 laser device within normal operation within it's enclosure).

Electronics Architecture

The electronics subsystem fulfills (E0: synchronized stage and laser control, E1: PCB integration).

Major components:

- Raspberry Pi Pico microcontroller
- motor driver PCB
- encoder interface board
- laser driver circuit

Power rails:

- 12 V motors
- 5 V laser
- 3.3 V logic

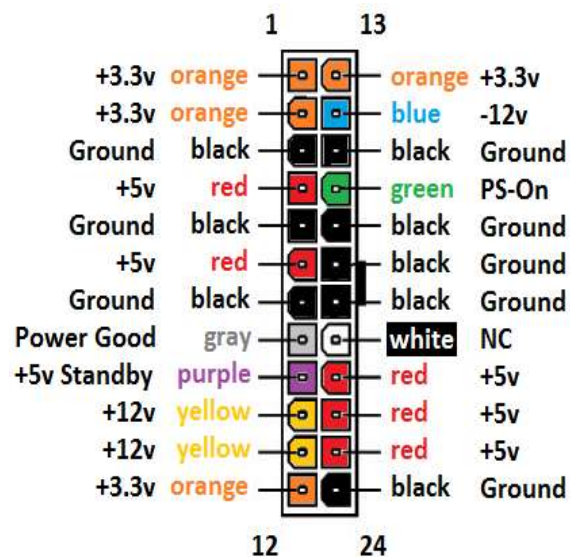


Figure 47: DELL ATX PSU pinout used for power-rail mapping in the electronics and safety design (Responsible: Armin — power, electrical interlock, and safety design).

The following pages outlines a number of schematics for the wiring of our components, including a simplified overview of the control system

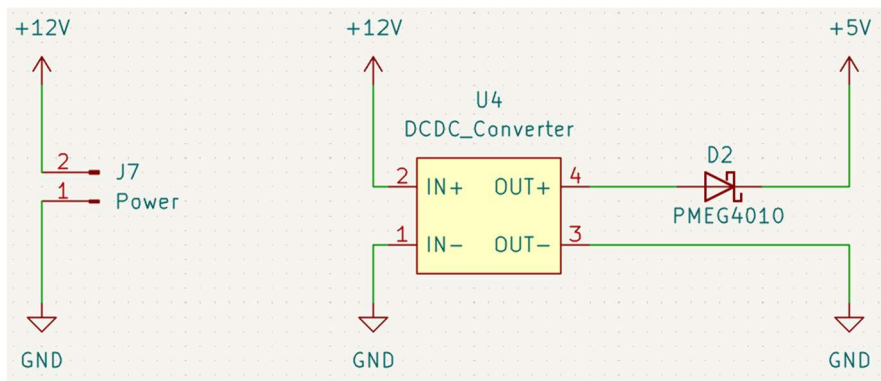


Figure 47: DC-DC Converter using the MH-MINI-360 step down buck converter board. The diode's purpose is to prevent reversing current into the converter, causing damage or unintended operation.

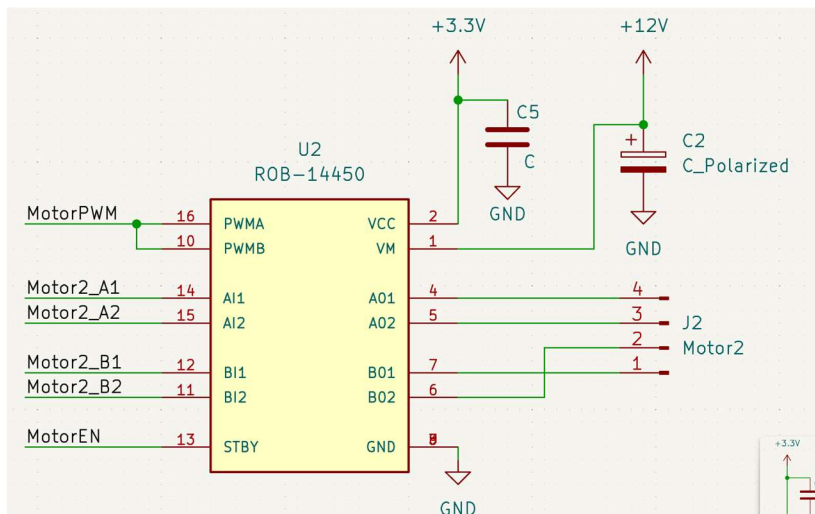


Figure 48: Motor driver wiring with connections to the Pico 2, with differentiating power supply for logic (3.3V) and power (12V). There are three of these. It's recommended to place decoupling capacitor (C5) on the logic supply for instant current when internal logic switches, and a 47 μ F-220 μ F electrolytic is recommended for stabilizing the motor power supply.

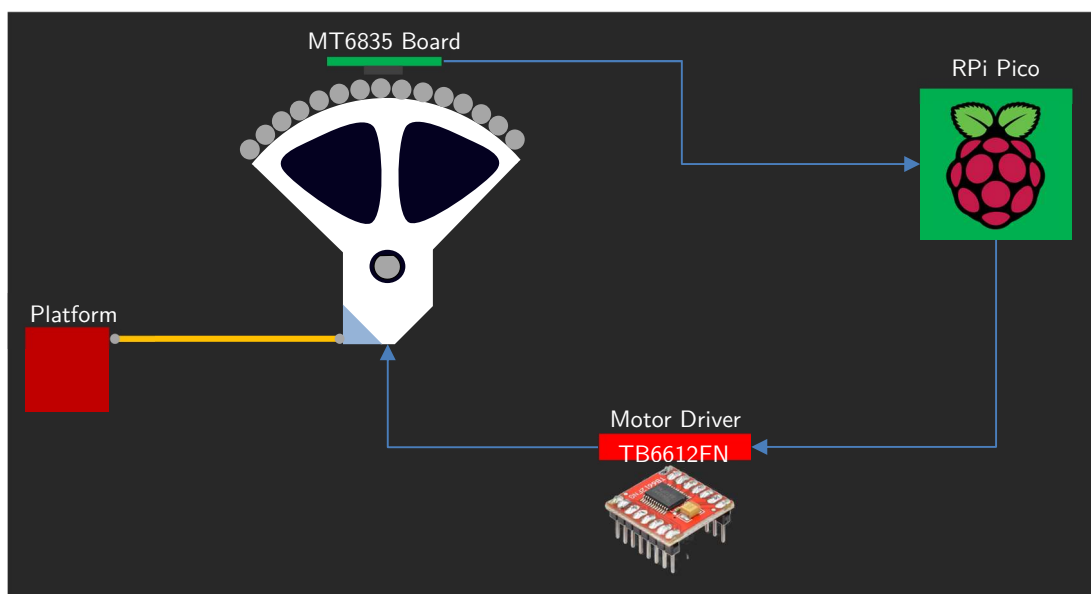


Figure 49: Left: Simplified overview of the closed-loop control system implemented on the firmware (stepper motor hidden). Right: MT6835 board used against the array arc of magnets.

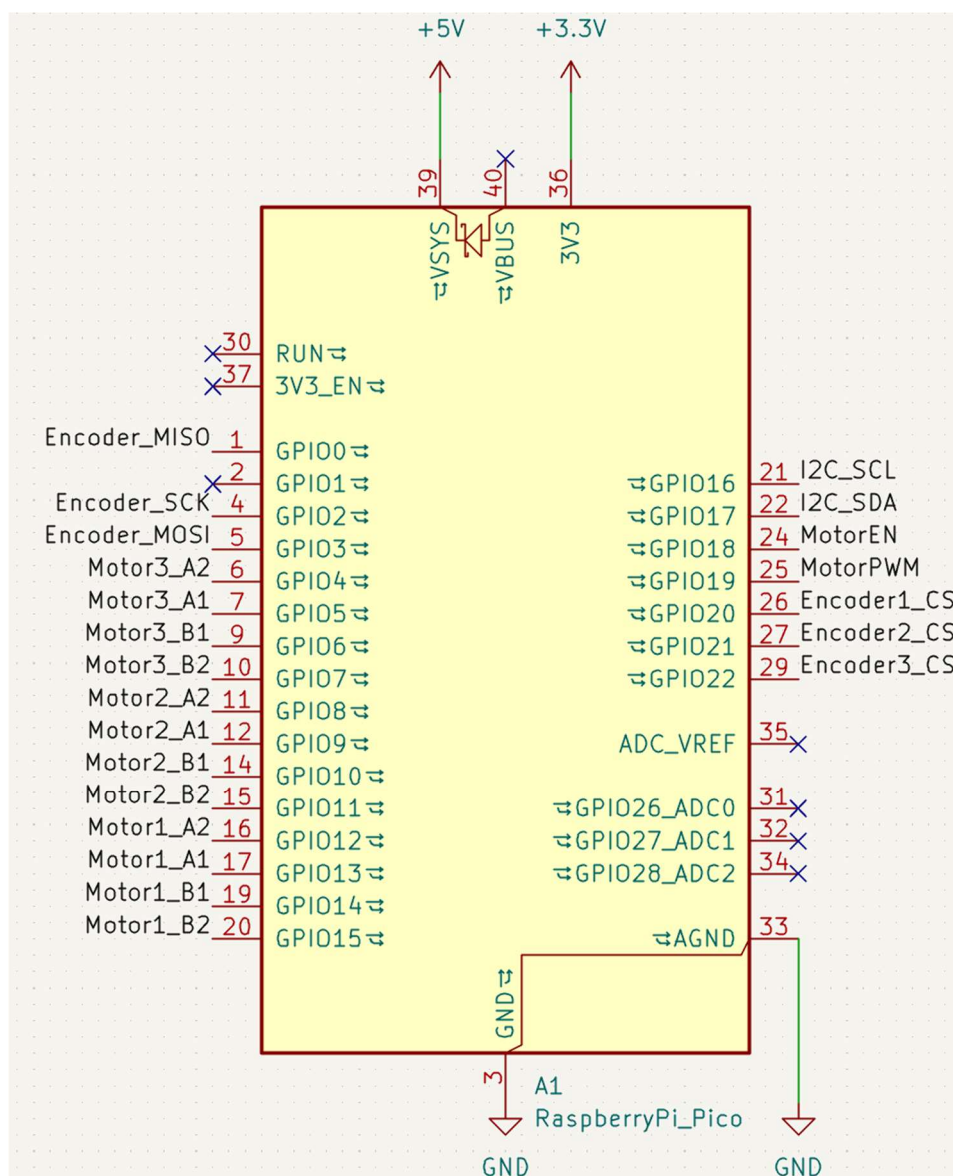
Current Feedback Control Loop (Implemented)

This specification fulfils: M0,M5

The current stage controller uses a cascaded feedback structure consisting of an outer position PI loop and an inner velocity PI loop, followed by a torque command path and PWM motor drive. Rotor angle is measured using the magnetic encoder and magnet array and stored in a calibration LUT. The calibrated rotor-angle signal is reused for both position feedback and velocity estimation.

This implementation provides very fine resolution because of the magnetic encoder and magnet array, but its absolute accuracy is not sufficient for the target positioning range of 300 nm to 750 nm defined in the performance requirements. To meet this accuracy goal, an additional position sensor must be introduced.

For the second sensor, the main options considered were a laser interferometer and a linear optical encoder. A laser interferometer was selected because the linear optical encoders



surveyed were typically specified to about $1\text{ }\mu\text{m}$ accuracy, which is not sufficient for the intended performance target.

In the following control-loop discussion, the laser interferometer is treated as a black-box measurement device that returns only position.

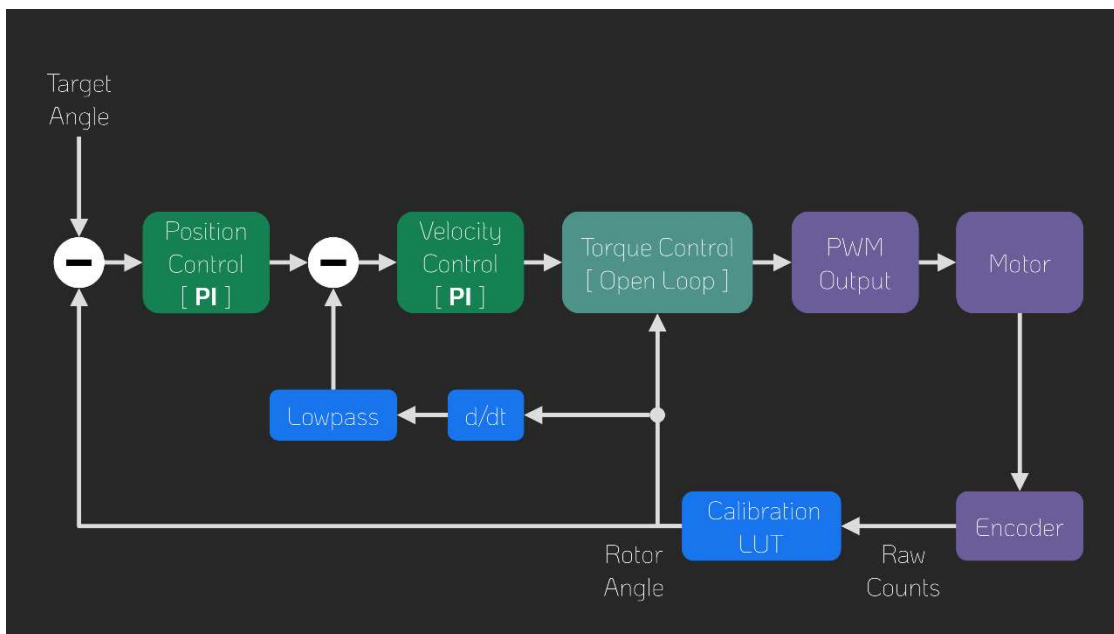


Figure 51 Implemented feedback control loop for the XYZ stage using the magnetic encoder and magnet array only.

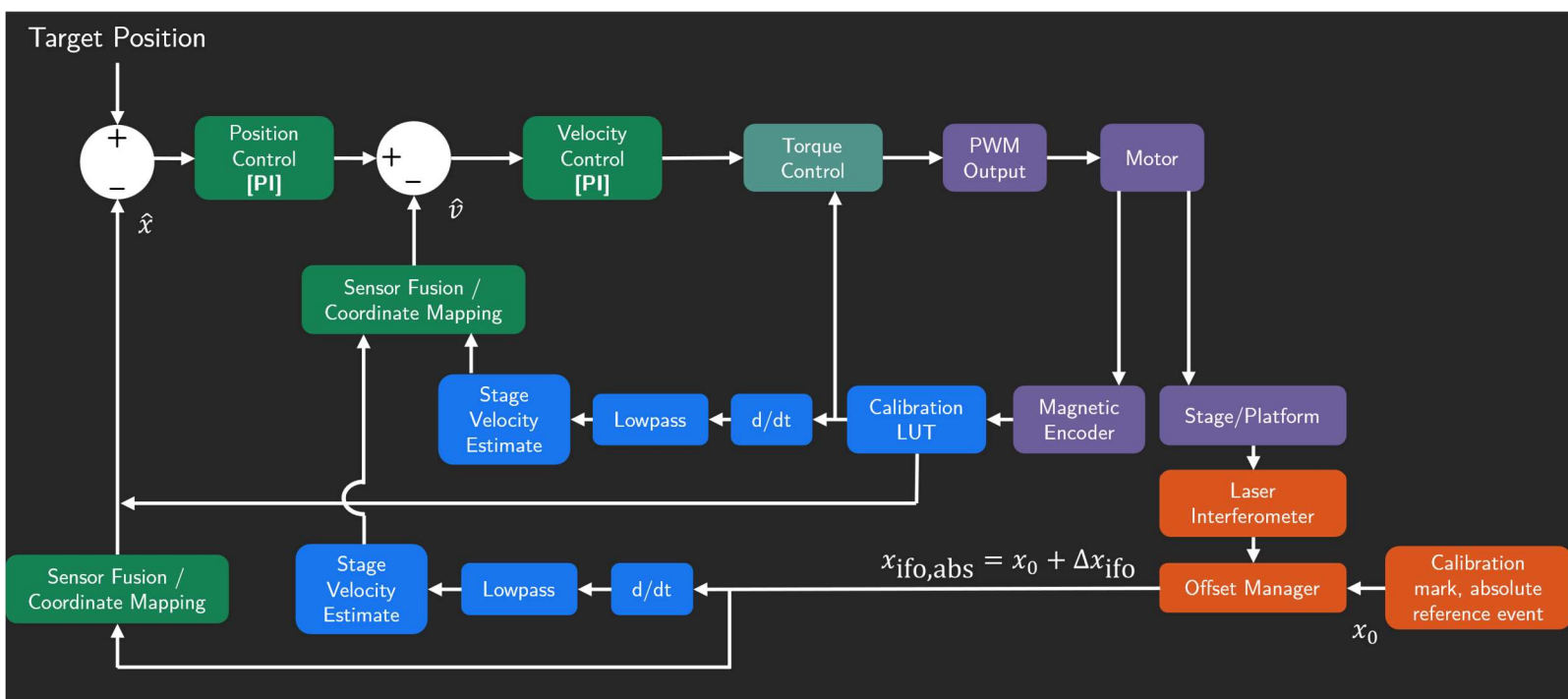


Figure 52: Laser interferometer integrated control loop

The interferometer is used as a high-accuracy relative displacement sensor rather than as a standalone absolute position sensor. Absolute stage position is established through a reference event such as the calibration mark. The stage position estimate is then formed by combining the stored absolute reference offset with the interferometer-measured relative displacement.

Sensor fusion requires that both sensors be expressed in the same stage coordinate system. For the magnetic encoder, this is achieved by mapping motor rotation into an estimated stage displacement using the motor-to-stage kinematic model and a calibration LUT. The interferometer provides the stage's relative displacement along the measurement axis, which is converted into an absolute stage-position estimate using the stored reference offset.

After this coordinate mapping, the two position estimates are fused using a complementary filter. The interferometer contributes the low-frequency, high-accuracy stage position information, while the encoder-derived estimate contributes the high-frequency motion content needed for responsive control.

The fused position estimate is

$$\hat{x}(s) = \frac{\omega_c}{s + \omega_c} x_{ifo,abs}(s) + \frac{s}{s + \omega_c} x_{enc}(s)$$

$x_{ifo,abs}$: Interferometer measured absolute stage position

x_{enc} : Magnetic encoder-derived estimate of stage position obtained from motor angle through the kinematic mapping and calibration LUT

The interferometer-based absolute stage-position estimate is formed as

$$x_{ifo,abs} = x_0 + \Delta x_{ifo}$$

x_0 : Absolute reference (calibration mark)

Δx_{ifo} : Change in interferometer measured stage position

Magnetic Encoder and Magnet Array

This specification fulfils: M0,M5

The XYZ stage uses a magnetic encoder and a magnet array to estimate motor angle position.

We use the MT6835 magnetic encoder. It has a 21-bit resolution. It's designed to measure the angle of rotation between the north and south poles of a magnet. Although you could place a single magnet on the motor shaft and measure its rotation directly, but in our case it does not provide enough effective resolution for sub-micrometer linear motion.

To counter this, we use a magnet array, which is alternating magnets in a linear row (Figure 50.1). This allows us to use a larger radius so the same angular change produces a longer arc length, which gives us a higher mechanical resolution. As the encoder moves along the array, each north-to-south transition appears to the sensor like a full magnetic rotation cycle. (Figure 50.2)

The optimal position to place the magnetic encoder is 4-6mm above the center of the array (Figure 50.3)

This method of motor angle sensing and the high resolution of the sensor gives us sub micrometer resolution capability. Unfortunately the magnetic sensor has an inherent limitation in it's accuracy, the factory calibration gives us $\pm 0.5^\circ$ of accuracy. [6]

Doing some math, with the motor horn 6mm away from the center of the motor shaft

$$\frac{0.5^\circ}{360} \cdot 2\pi(6mm) = 52um$$

So the magnetic encoder is limited to giving $\pm 52um$ of accuracy at the motor horn. This isn't even factoring in the translation from motor horn to end effector/ platform movement. But it clearly isn't accurate enough for our specs.

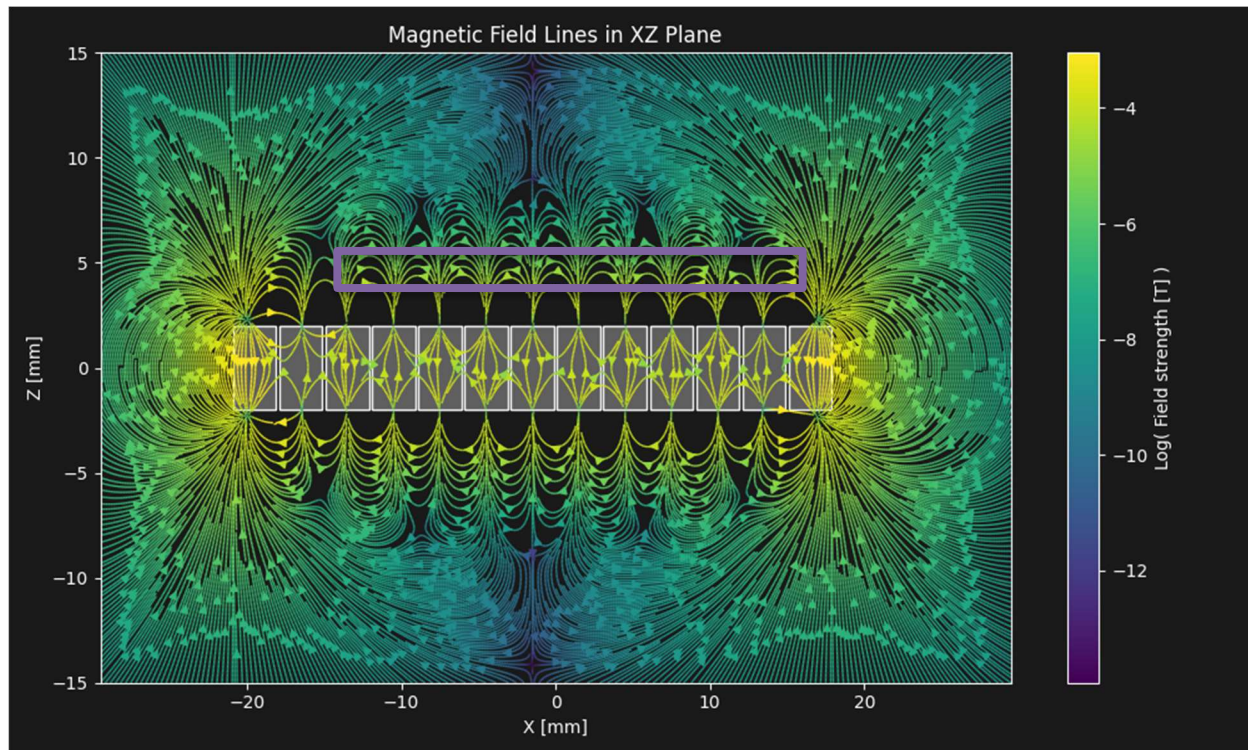


Figure 53.1 Magnetic Field Lines Simulation of magnet array in the XZ Plane



Figure 53.2 Zoomed in View of the Magnetic Field Lines Simulation in the XZ Plane

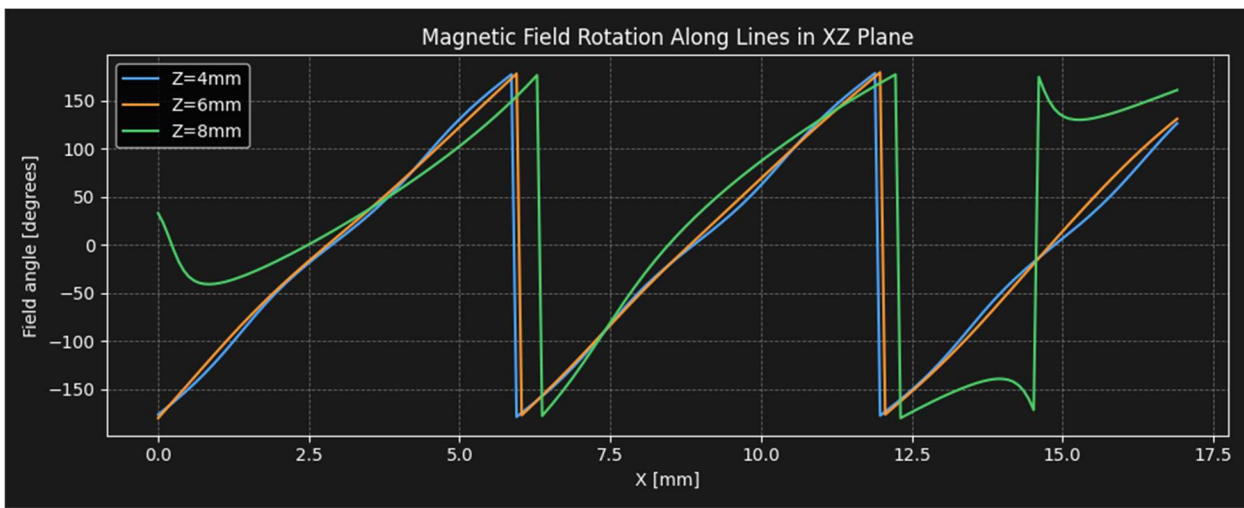


Figure 53.3 Magnetic Field Lines Rotation Along lines in XZ Plane

Laser Interferometer

This specification fulfils: 5.1: XYZ stage accurate to 300nm - 750nm

There are many designs to a Laser Interferometer, we are going to use the Michealson interferometer. A single laser beam is divided by a beam splitter into a reference arm and a measurement arm. The two beams are reflected back to the beam splitter and recombined at the detector. The recombined intensity depends on the optical phase difference between the two arms, so motion of the measurement arm is converted into a periodic intensity signal that can be counted and interpolated to recover displacement.

The resulting light intensity can be detected by a photodiode, and converted to phase using this equation

$$P_{det} = P_0 \frac{1 + \cos(\Delta\phi)}{2}$$

Because the light travels to the moving reflector and back, the optical path changes by

$$\Delta L = 2x$$

So the corresponding phase shift is

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta L = \frac{4\pi x}{\lambda}$$

So one full interference cycle occurs when the moving arm changes by

$$x = \frac{\lambda}{2}$$

Each full oscillation of the interference signal corresponds to a displacement of one-half wavelength of the laser. So we count the number of fringes, N

$$x = N \frac{\lambda}{2}$$

For phase interpolation to get even higher accuracy, we use

$$x = \frac{\lambda}{4\pi} \Delta$$

This is the basic theory of how the interferometer works. The implementation is as follows.

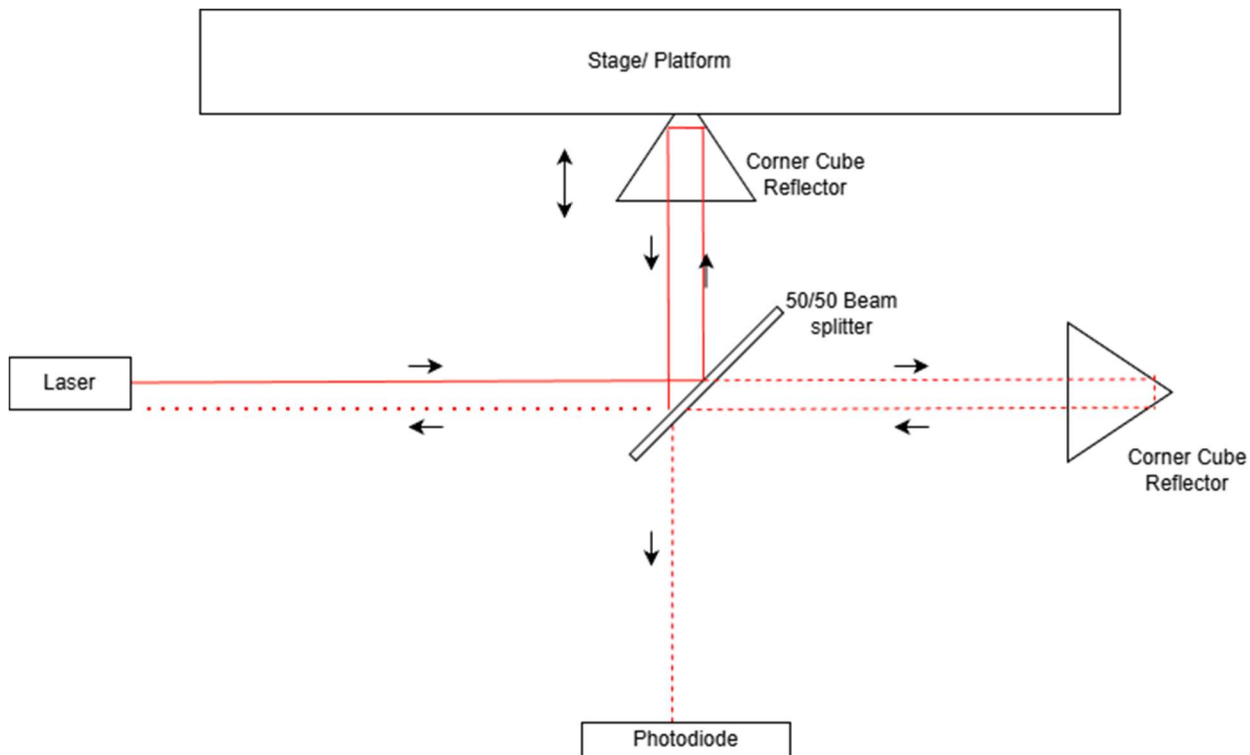


Figure 54.1 Michealson Interferometer using 2 corner cubes

Corner cubes are used for the reflection because of alignment robustness. The corner cube reflects the laser parallel and opposite to the incident beam.

To be able to see the fringes, coherence length needs to exceed the arm path-length difference, in this case its $\sim 2 \times$ measurement field. Since we want to measure the movement of at least X and Y axis of the platform, and the platform needs to move 30mm, we need a coherence length of at least 60mm. A cheap laser diode typically has a coherence length of μm which is far from 60mm. So we will apply a few techniques to boost the coherence length of the laser.

Temperature and Current Control

Tight temperature control along with stable drive current reduces the wavelength drift.

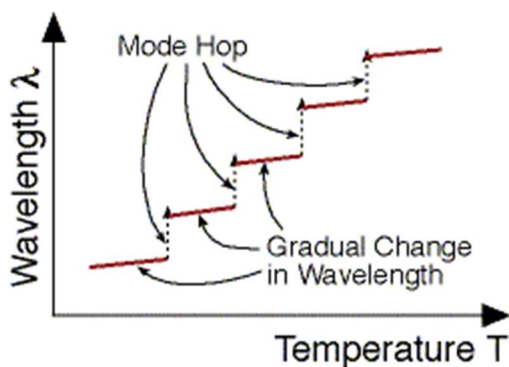


Figure 54.2 Wavelength vs Temperature of a laser diode's output. Reproduced from [5]

You can see in Figure 51.2, The change in temperature is directly correlated with the wavelength. For a good coherence length we want to stay within one wavelength only

because wavelength is directly correlated with coherence length. A lower change in wavelength means a longer coherence length.

$$L_c = \frac{c}{\Delta f} ; f = \frac{c}{\lambda} ; \Delta f = \frac{c}{\lambda^2} \Delta \lambda$$

L_c : Coherence length

Δf : Change in frequency

So our goal is to stabilize the temperature in one of those stable modes.

Ultra-stable current driver

An ultra-stable current driver would have very low ripple and low noise. Injection current fluctuations affect the laser's optical frequency and can push the device into an unstable mode or other wavelength. So we want to keep the ripples and noise to a minimum to stay within the mode we aim for, keeping the wavelength relatively constant.

Single-mode operating-point search

Some temperature/ current combinations are stable while others produce mode hopping. So we have to scan the current and temperature of the laser and observe the fringes we get from approximately equal length arms (fixed and moving arm). We aim to find the point of operation that the fringe visibility is maximized, and intensity noise is minimized.

Prototype Resolution vs. Accuracy Observation (Review Meeting 2)

As an initial prototype test, we used an already fabricated SFU wafer feature and targeted motion across the inside width of the letter U (nominally 150 μm). The GUI jog step was set to 1 μm , and traversing that distance required 110 clicks.

This result indicates good motion resolution (small controllable increments are available), but poor absolute accuracy (commanded displacement and true displacement are not yet well matched). Improving absolute positioning accuracy is therefore a major Capstone B priority.

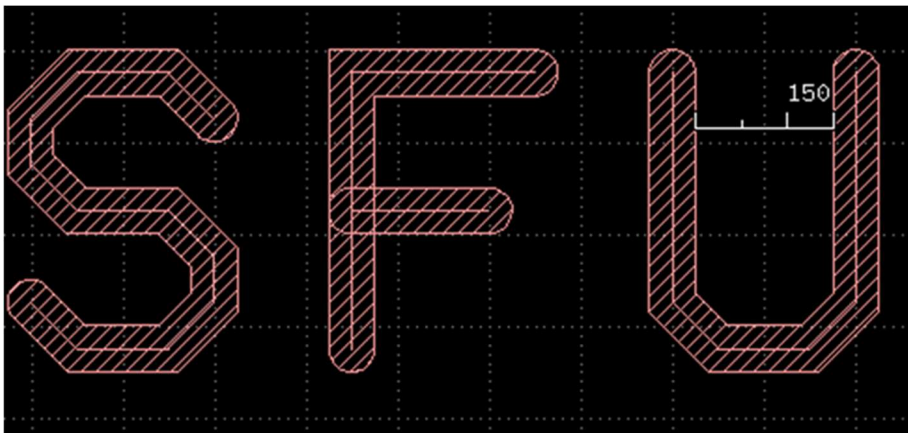


Figure 55.1: Reference SFU geometry used for the prototype motion check; the U-width target is marked as 150 μm .



Figure 55.2: Live camera/GUI view during the motion test, showing the SFU feature under the software crosshair.

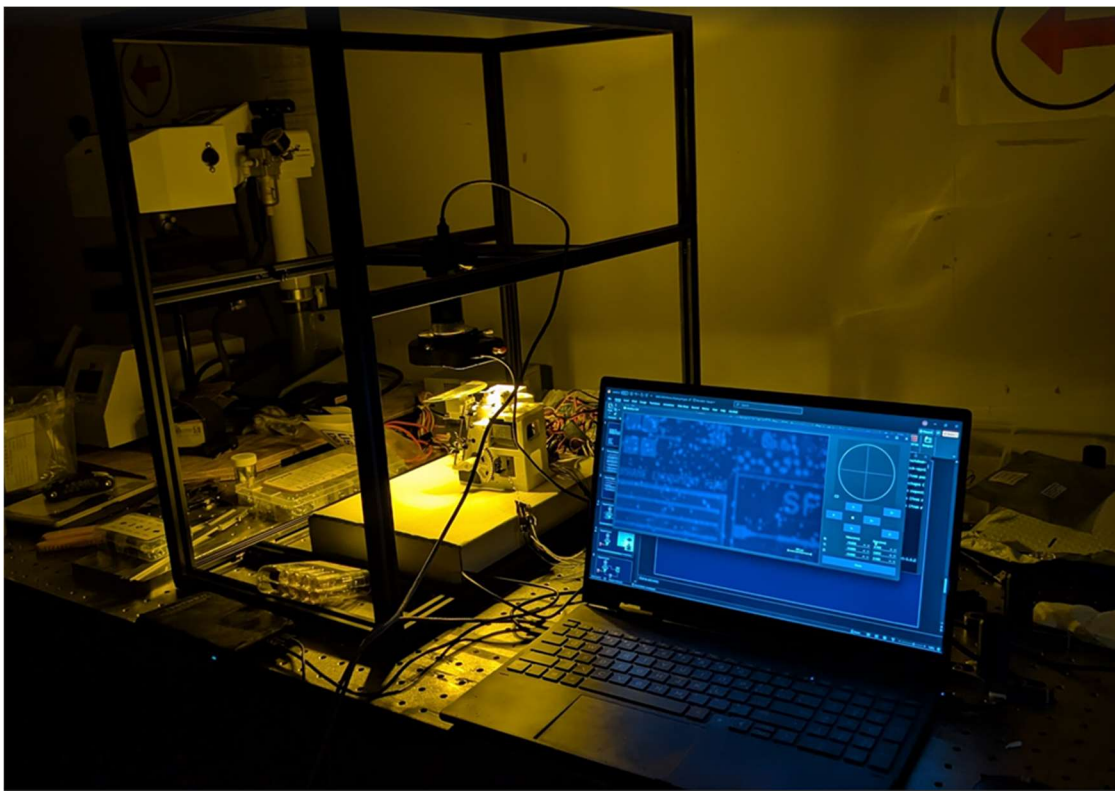


Figure 55.3: Physical setup used for the Review Meeting 2 live demo of electronics/firmware-controlled stage motion and GUI feedback.

Key ideas to improve accuracy in Capstone B:

- Calibrate axis scale factors ($\mu\text{m}/\text{step}$) using known metrology features and store corrected conversion constants in firmware.
- Build a bidirectional correction table to compensate backlash, hysteresis, and direction-dependent gain error.
- Add a laser interferometer as absolute outer-loop position feedback while retaining encoder feedback for high-bandwidth commutation/control.

- Upgrade control tuning (feedforward, anti-windup, and friction compensation) to reduce steady-state error after each jog.
- Add thermal warm-up and periodic re-zero routines to reduce drift over long exposure sessions.
- Add an automated GUI calibration routine that measures known fiducials and applies correction before writing. Perhaps take advantage of computer vision techniques to see how "off" accuracy is (though that could be very slow processing)

Construction Sequence

The full device is assembled in the following order.

1. Construct aluminum frame
2. Install XYZ stage
3. Mount optical components
4. Install camera system
5. Install electronics
6. Assemble enclosure panels
7. Install HEPA filtration system

Appendix A: Sustainability and Safety

Issues related to sustainability and safety are evaluated across the full product life cycle, including sourcing, assembly, operation, maintenance, upgrade, and end-of-cycle recovery.

A.1 Cradle-to-cradle Life cycle analysis

1. **Material sourcing:** We're building the system from recovered/salvaged components wherever safe and technically possible, such as used aluminum plates, used objective lenses, aluminum stock and power components from decommissioned fab equipment as well as a previously used image sensor. To whoever builds a similar machine on a budget, we recommend the same approach in hopes to reduce e-waste and other functionally good materials from ending up as garbage. Electronic Components such as the encoders and motor drivers can be easily found at vendors such as DigiKey, Amazon, and AliExpress.

2. **Assembly:** we favor recyclable structural materials such as aluminum and want to keep our CNC design small to avoid waste. In theory, most of our 3D prints can be recycled for example with SFU EWB's 3D print filament recycling project, which mainly uses bottles but could also use 3D prints. We also prioritize making all 3D printed component modular and swappable so that there is no need to throw away entire 3D prints just because they no longer serve our specific design.

3. **Use phase:** we will operate laser outside of an enclosure only within the printer room whilst using protective eye goggles. When moving out of the printer room, it will be when the enclosure is done and approved by SFU Laser Expert Shawn Sederberg. It will of course operate with an interlock safety control, and localized clean airflow to avoid possible contaminants arising when the laser fires.

4. **Upgrade/repair phase:** We will explore different ways the machine could fail during use and offer accessible ways for repairing and replacing parts. Our modular design choices will allow part replacement rather than full-system disposal.

5. **End-of-life phase:** We want to allow the end user to upgrade their system rather than dispose of it if say the laser diode dies out or if the XYZ stage no longer provides resolution relevant for research. If the user suddenly wants to use a 255nm laser instead of 405nm, it could be upgraded and it could give them better feature size instead of having to buy a brand new machine. In the case that the machine really has come to the end of its life, we will allow an easy way to disassemble by module to recover aluminum, electronics and reusable optics.

A.2 Design choices for sustainability

The current design already includes several sustainability-positive choices:

- Extensive use of salvaged electrical/safety parts where performance and safety are verified.
- Use of magnetic encoders rather than full gearboxes, generating far less material in total that could end up in the garbage while using encoders that could be reused for future projects.
- Use of aluminum stock and aluminum sheet structures that are durable and highly recyclable.
- Use of donated objective lenses/optics to reduce procurement footprint and extend component life.
- Modular frame and electronics architecture that supports repair, swap-in upgrades, and lower lifecycle waste.
- Reuse-oriented prototyping strategies that delays custom fabrication or orders until requirements are validated.

A.3 Safety Integration Across the Lifecycle

- Interlock-driven laser shutdown and emergency-stop power isolation.
- Safety measures in the printer room (hazard signage, controlled access, locked door, and operating procedures) to reduce risk of harm to bystanders. This is verified by SFU laser safety expert Shawn Sederberg.
- We will repeat safety verification after any hardware changes during iterative design.

Appendix B: Testing Plan

This section expands the acceptance testing strategy from the requirements specification into measurable subsystem and component checks for Capstone B.

Testing is executed in gated stages so that unsafe or low-confidence states are caught early. We will always begin any demo session with the following steps to demonstrate each sub-system.

1. Safety pre-checks with proper protective gear (does the interlock work? Does the emergency stop button work?) – This verifies the electrical subsystem
2. Motion test and validation (what is the resolution, accuracy, repeatability) just by looking at a premade wafer, together with good responsiveness and control with the GUI – This verifies the mechanical, controls and software subsystems.
3. Optical exposure validation (spot, focus, area of wafer that can be covered) and later on successful exposure of a pattern – This verifies the optics subsystem and on a high level the controls, mechanical and software subsystems.
4. Pattern-writing progression from simple (lines, rectangles) to application-specific geometries (circles, inductor swirls, Schottky diode thin zig-zag shapes...) – This verifies the higher complexity that the software subsystem is responsible. The more complex shapes we can demo, the better.
5. Demonstration of the system that the system can repeat the same results many times without failure while being operated by the end user – Once the end user is free to use the machine to their needs, we may come to a point where the project is usable for the Schottky diode fabrication.

Notes about the Class 2 Laser and safety verification

All team members working on the project have undergone or are undergoing SFU's laser safety program, which will include an in-person training session in April. Two members already have laser safety certification and will be the two primary members who will run any laser tests and experiments in the printer room. Because the prototype is operated as a Class 2 visible-laser system during alignment/testing, the following checks are required before exposure runs:

- Verify the laser module output and mode are consistent with Class 2 operation assumptions (nominally 1mW visible beam) but treat as if it's a Class 3 beam.
- Verify door/flap interlock and emergency-stop chain produce an immediate (<1 ms) power halt to the laser and the XYZ stage drives.
- Verify beam containment controls: enclosure closed, protected viewing conditions, and warning signage active. For Capstone B, it'd be worthwhile to have a "Laser ON" indicator designed with an LED or even an LCD display.
- Verify emergency-stop reset behavior requires intentional and safe user action before laser re-enable. Verify that the system can resume normal operation after a reset.

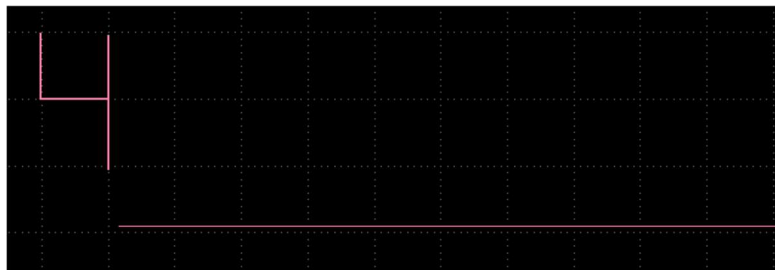
Subsystem Acceptance Matrix

Test	Requirement	Metric/Model	Initial Acceptance Target
Spot size verification	E4.1, E4.2	Microscope-derived line/spot measurement	3 μ m feature size capability
Stage repeatability	M5.2	Return-to-point error over repeated moves	0.3 μ m to 0.7 μ m repeatability
Stage absolute accuracy	M5.1	Commanded vs. measured displacement error	0.3 μ m to 0.7 μ m accuracy
Camera-to-laser alignment	M4.6, S4.5	Crosshair-to-exposure offset	Within acceptable range for multi-layer alignment
Air cleanliness check	M7.1	Local particle count trend near wafer area using a borrowed high-end particle counter	Meets planned local ISO Class 5 operating target
Laser safety test	G1.1, G1.2	Interlock and E-STOP button test	Verified hard shutdown and controlled reset

Testing the performance of our system

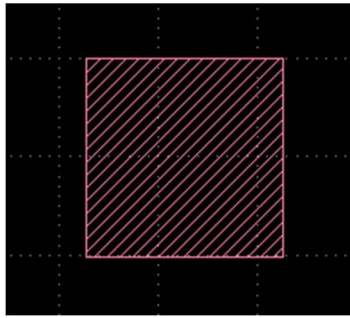
Pattern validation is intentionally staged from low-complexity which will be demonstrated in Capstone A to high-complexity which will be optimized throughout most of Capstone B, using self-made .GDS files as well as ones provided by Michael Adachi's graduate students.

1. Simple line test: verify line continuity, width consistency, and endpoint placement error.

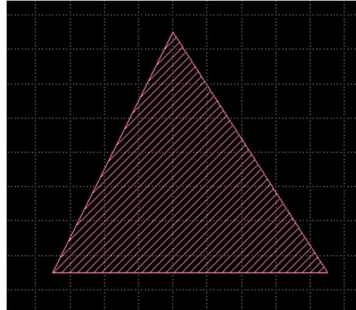


2. Unfilled square test: verify X and Y two-axis movement

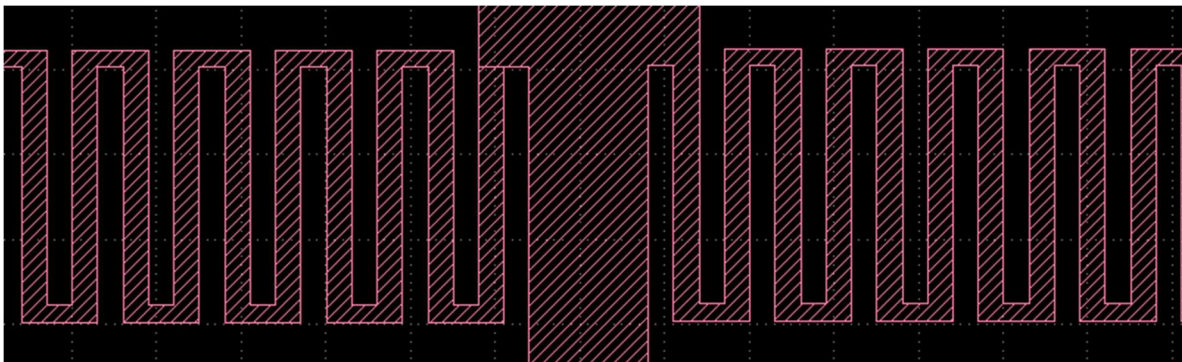
3. Filled square test: verify ability to properly fill shapes



4. Triangle test: verify diagonal movement rather than just straight X and Y movement



5. Circle test (optional): circles on an XY stage are very difficult to do. Circles are not needed for the Schottky diode design but prohibit certain applications by researchers, for example who wish to make inductors or specific MEMS devices.
6. Schottky diode snippet test, which has many complicated zig-zag shapes and patterns.



By the 6th test, we can almost verify that our machine fulfills the requirements for our Capstone B goal of fabricating the Schottky diode. Certain tests will fail due to certain constraints by different subsystems, perhaps the control system firmware needs to be improved to improve accuracy, perhaps the mechanical constraints play a role, perhaps the software laser writing algorithm, perhaps the power used by the laser isn't clean enough and generates fluxes in the intensity and therefore exposure uniformity... and so on. This is a way to validate each sub-system by debugging what the issue at hand traces to which subsystem.

Data collection & Pass/Fail Rules

We will use collected microscope images, GUI and firmware logs, commanded G-code files, for each test run. We will run tests at least three times to separate one-off success. We will classify the following:

Pass: A shape/pattern is successfully developed and seen on the microscope to reasonable accuracy with minimal artifacts and dust particles present.

Fail: the shape is not close or coherent enough to the design made. From then on, we will do debugging to see what the issue is at hand (mechanical, firmware, algorithm, electrical...)

Appendix C: Engineering Standards

The project aligns with recognized standards as design guidance targets. We do not claim formal certification, but we do follow relevant standards and factor them into our design choices.

C.1 Ten Key Engineering Standards

1. **IEC 60825-1** (laser product safety classification and requirements): supports laser classification, labeling, and protective control strategy.
2. **ANSI Z136.1** (safe use of lasers): supports operating procedures, hazard controls, and design for a Class 1 enclosure.
3. **ISO 12100** (machinery risk assessment and risk reduction): supports systematic hazard identification and mitigation. Including the XYZ stage in the interlock will ensure no one jams their fingers in the mechanical moving parts of the XYZ stage.
4. **IEC 60204-1** (electrical equipment of machines): supports emergency-stop behavior, protective bonding, and machine electrical safety architecture.
5. **IEC 61010-1** (laboratory electrical safety): supports safe bench-top integration and wiring practices.
6. **IEC 61326-1** (EMC for measurement/control/lab equipment): we are looking into how to isolate our magnetic encoders from receiving interference from the environment.
7. **ISO 14644-1** (cleanroom air cleanliness classification): supports local ISO Class 5 cleanliness target definition, ideal cleanroom cleanliness for research-grade fabrication.
8. **ISO 14644-2** (cleanroom monitoring): supports periodic cleanliness verification and trend tracking during operation.
9. **IPC-A-610** (electronic assembly acceptability): supports PCB/assembly workman-ship quality and maintainability as we are including mainly headers on our PCB, making it easy to swap electronic boards/chips that may have fried. In the unlikely event that the PCB fails, one could order a new one easily, or use one of the ten that we originally ordered (minimum order 10)
10. **ISO 14044** (life-cycle assessment framework): supports cradle-to-cradle sustainability analysis and material recovery planning.

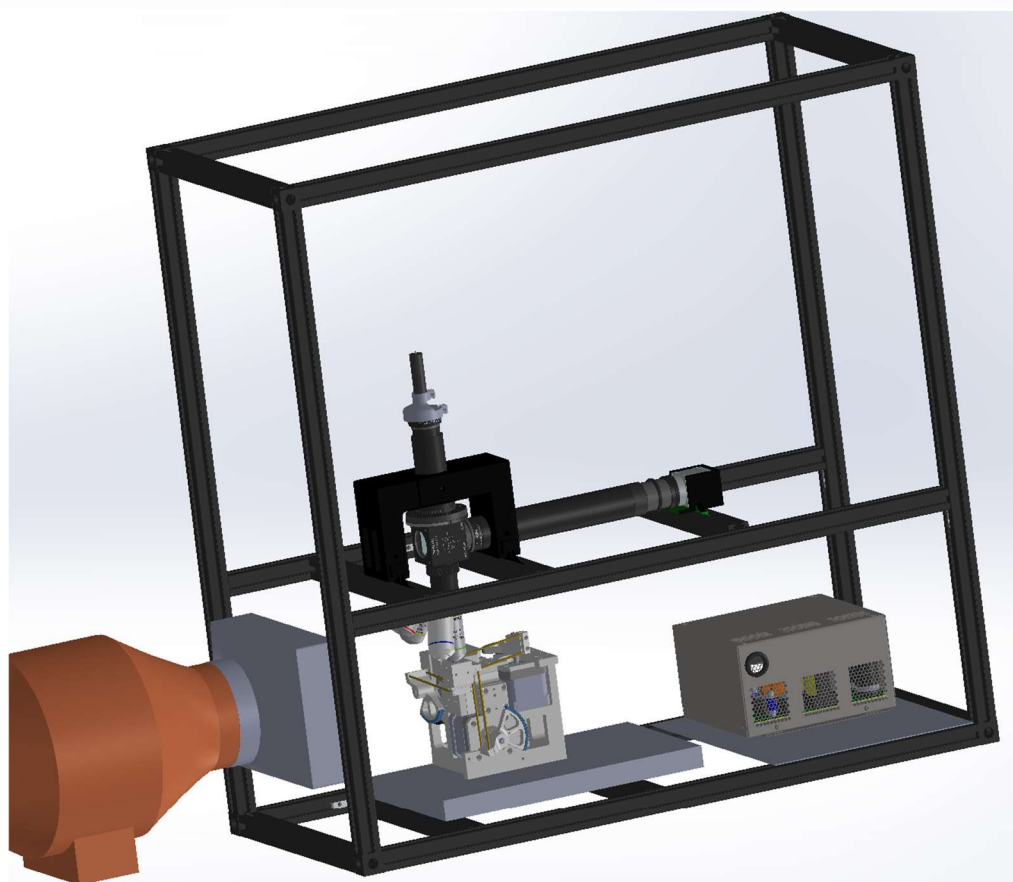
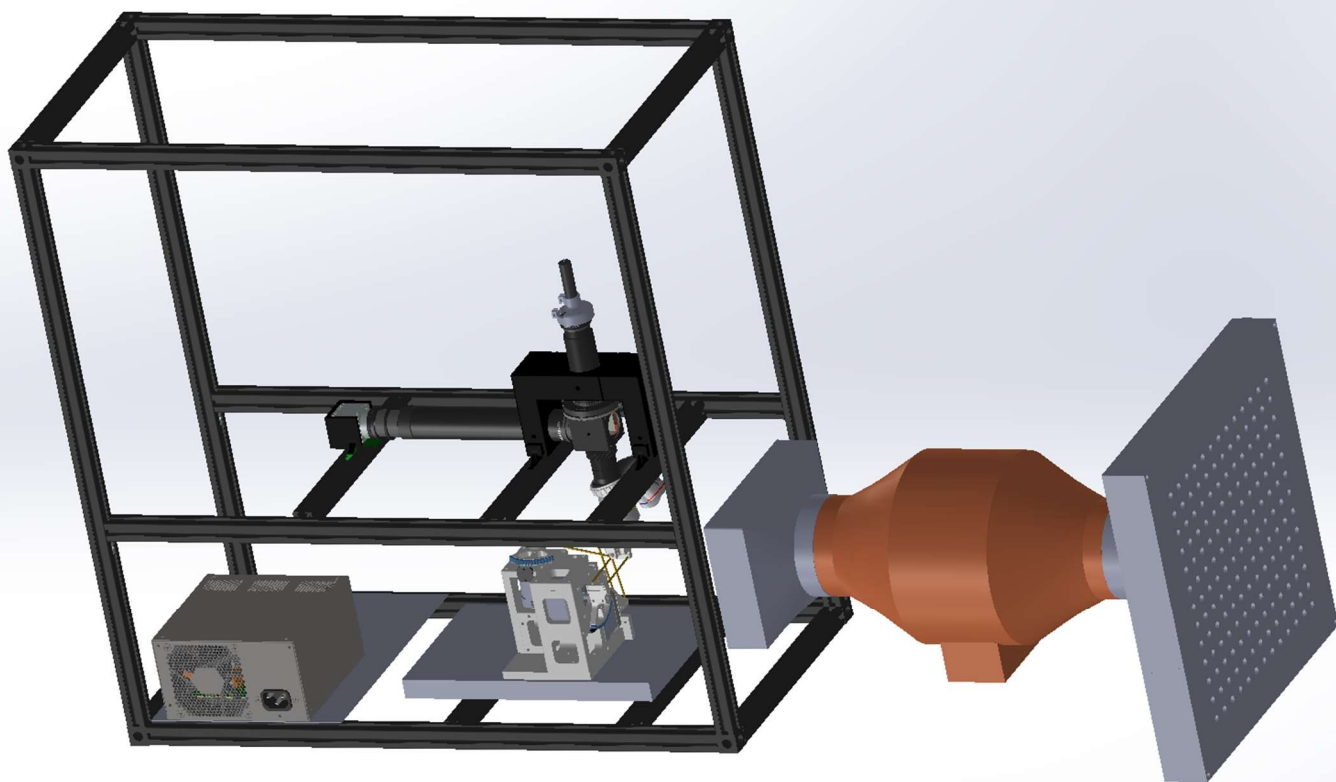
C.2 Five standards for user interface and device interaction

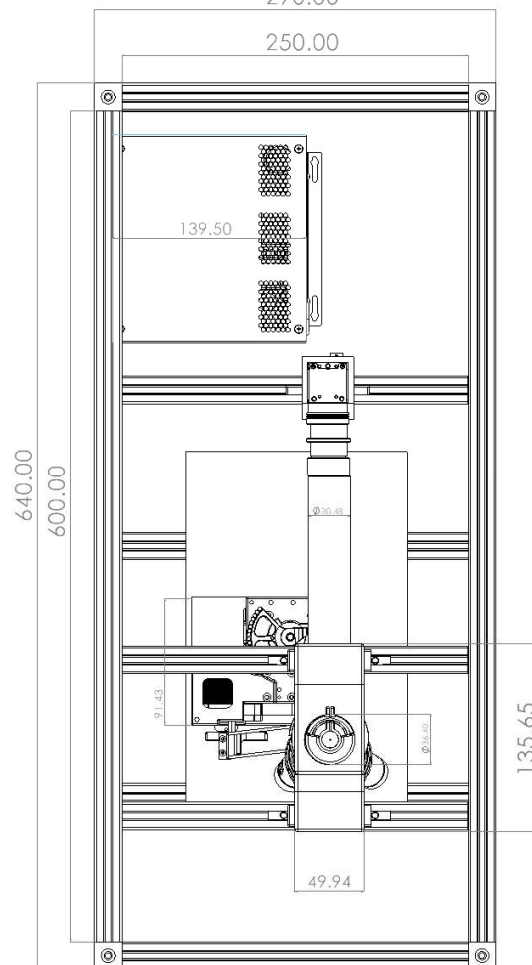
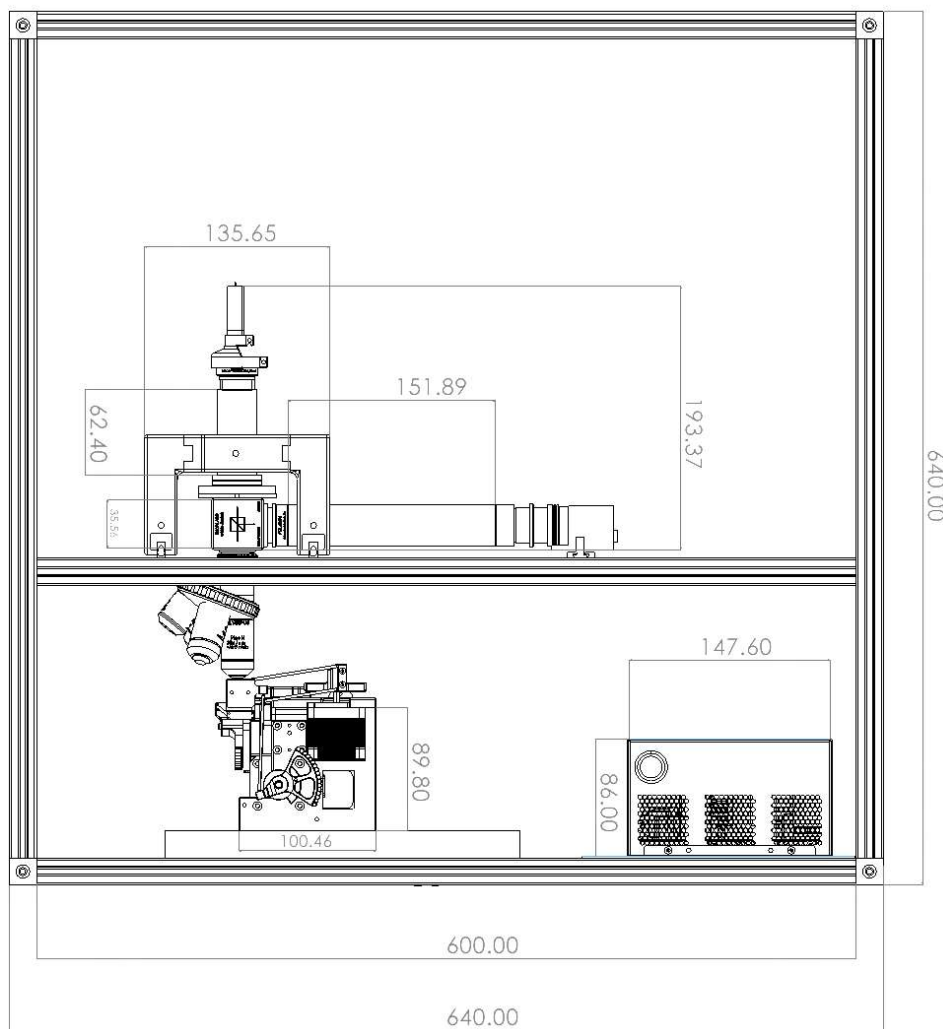
1. **ISO 9241-110** (interaction principles): supports an easy to use GUI flow for faster alignment and reduced user burden.
2. **ISO 9241-210** (human-centered design): supports what the user is looking for when it comes to alignment, having room for human error with alignment
3. **ISO/IEC 25010** (software quality model): supports usability, reliability, maintainability, and cross platform and cross design software. We will look more into how to automate the alignment process.
4. **WCAG 2.2** (accessible interaction guidance): supports readable, operable controls and improved interface accessibility. With GUI libraries like GTK, it's possible to support multiple languages.
5. **ANSI Z535.2** (facility safety signage): supports clear user warnings and entry instructions around active laser operation areas.

Project decisions that strongly align with these interaction-focused standards include:

- modular architecture for easy repair, part swap-in, and future subsystem upgrades.
- streamlined alignment workflow to reduce operator time versus legacy mask aligners.
- modern GUI quality-of-life features (live view, responsive controls, crosshair alignment).

Appendix D – Full CAD Rendering and Drawing of the system





Conclusion

LithoNow provides a research-grade maskless lithography system capable of producing micro-scale semiconductor patterns without photomasks and without the need for a cleanroom.

The design integrates precision motion control, optical focusing, software automation, and environmental contamination control to satisfy the requirements provided by the Nano Device Fabrication research group.

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